AKTU B.Tech I-Year

FME

(Crash Course)

ONE SHOT Revision

Unit-5

Measurement and Mechatronics

By M S Tomar Sir

Unit-5: Introduction to Measurement and Mechatronics

Introduction to Measurement: Concept of Measurement, Error in measurements, Calibration, measurements of pressure(Bourdon Tube Pressure and U-Tube Manometer), temperature(Thermocouple and Optical Pyrometer), mass flow rate(Venturi Meter and Orifice Meter), strain(Bonded and Unbonded Strain Gauge), force (Proving Ring) and torques(Prony Brake Dynamometer); Concepts of accuracy, precision and resolution.

Introduction to Mechatronic Systems: Evolution, Scope, Advantages and disadvantages of Mechatronics, Industrial applications of Mechatronics, Introduction to autotronics, bionics, and avionics and their applications. Sensors and Transducers: Types of sensors, types of transducers and their characteristics.

Overview of Mechanical Actuation System – Kinematic Chains, Cam, Ratchet Mechanism, Gears and its type, Belt, Bearing.

Hydraulic and Pneumatic Actuation Systems: Overview: Pressure Control Valves, Direction Control Valves, Rotary Actuators, Accumulators and Pneumatic Sequencing Problems.

AKTU B.Tech I-Year
UNIT-5: Introduction to Measurement and Mechatronics
Most Important Questions

Q.1	Write short notes on (i) Measurements of pressure(Bourdon Tube Pressure and U-Tube Manometer) (ii) Measurements of pressure(Bourdon Tube Pressure and U-Tube Manometer) (iii) Measurements mass flow rate(Venturi Meter and Orifice Meter) (iv) Measurements strain(Bonded and Unbonded Strain Gauge) (v) Measurements force (Proving Ring) and torques (Prony Brake Dynamometer)
Q.2	Explain the various errors in measurement and the practices that are needed to minimize them. Also write difference between accuracy, precision.
Q.3	What are sensors and transducers? Enumerate the various types of sensors and transducers.
Q.4	Define mechatronics. Also write down the advantages, disadvantages and applications of mechatronics.
Q.5	What are Autotronics, bionics, and avionics? Also, write their applications. Also define Calibration.

AKTU : B.Tech (I,II,III-Year)

S.N.	Full Courses available in application		
1	CS, IT & CS Allied	COMBO (Odd + Even) SEM	COMBO Odd/Even SEM
2	EC & EC Allied		
3	ME & ME Allied		
4	EE & EE Allied		

Validity till semester last exam

Helpline No. 7819 0058 53

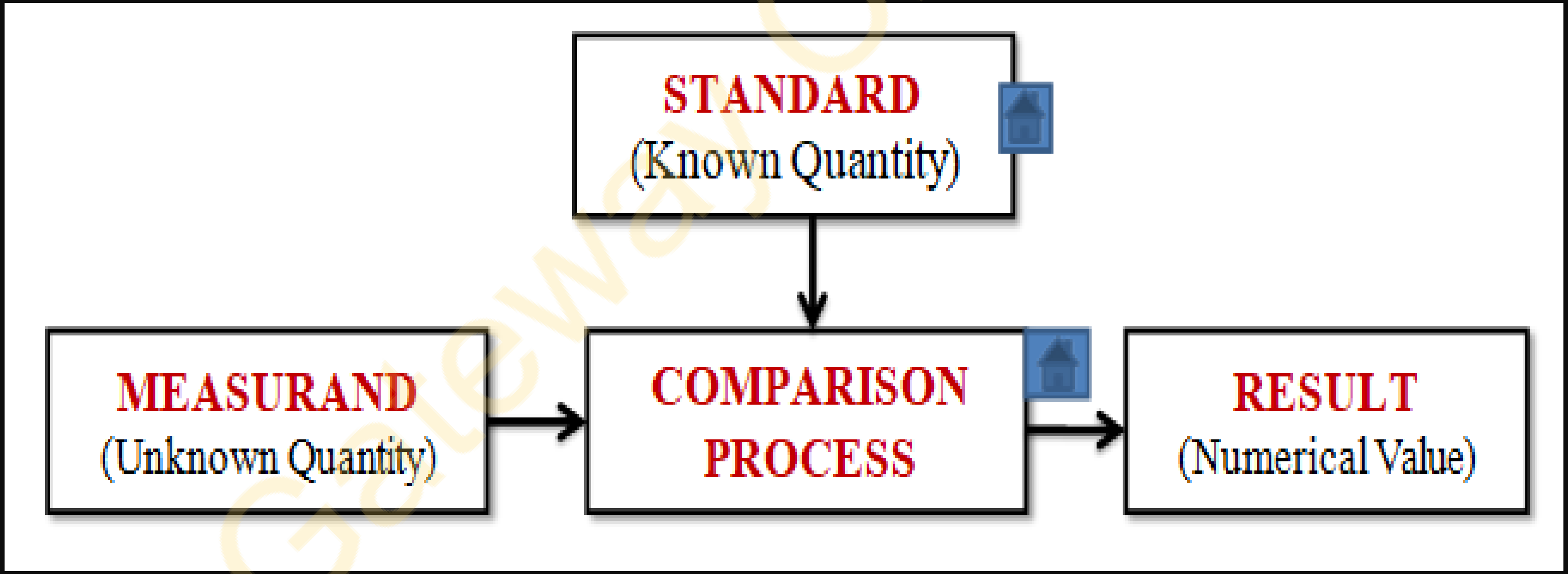
Full Course B.Tech Sem-II

MEASUREMENT

The measurement is the result of comparison between a quantity whose magnitude is unknown(known as measurand), with a similar quantity whose magnitude is known (known as standard) .

OR

It is the process of obtaining a quantitative comparison between a predefined standard and measurand (i.e the quantity being measured)



Measurand : A physical quantity such as length, weight, and angle to be measured.

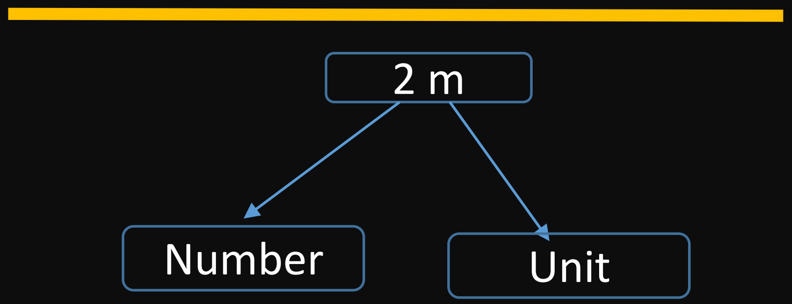
Standard/Reference: A physical quantity to which quantitative comparisons are to be made, which is internationally accepted.

- Measurements are always made using an instrument of some kind.

- Although the basic objective of a measurement is to provide the required accuracy at a minimum cost.

Result of measurement consist of two parts:

1. Number of measurement
2. Unit of measurement



In order for the results of measurements to be meaningful, there are some requirements

- The standard used for comparison purposes must be accurately defined and should be commonly acceptable.

- The standard must be of the same character as the measurement.

- The apparatus used and the method adopted for comparison must be Justifiable.

Example of measuring instrument or tool:

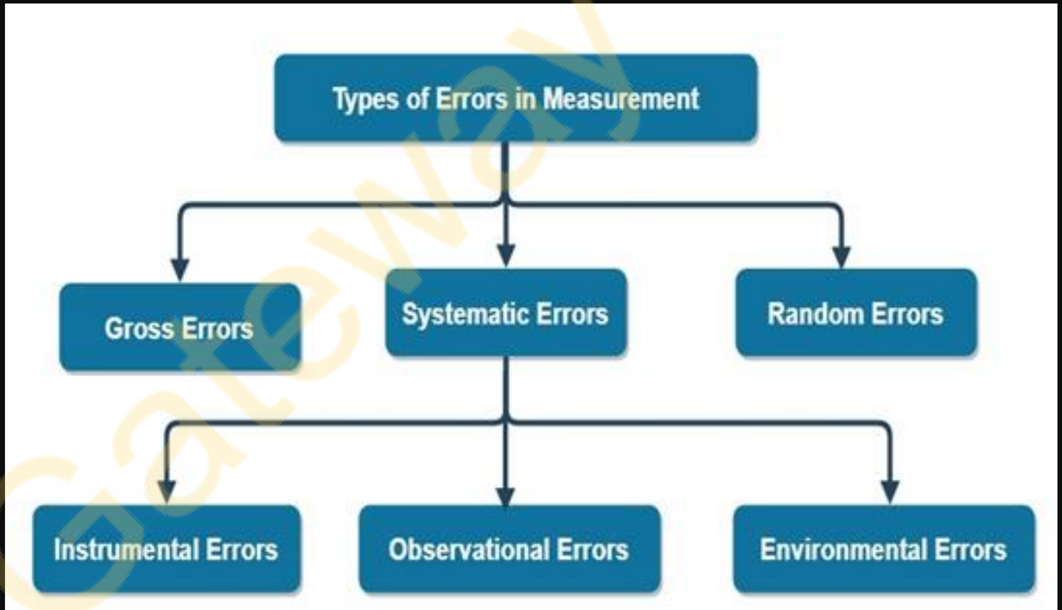
- Ruler
- Thermometer
- Stop watch or watch
- Weighing Machine etc

ERROR IN MEASUREMENT

It is defined as a difference between indicated or measured value and true value.

ERROR = TRUE VALUE – MEASURED VALUE

- No measurement can be made with perfect accuracy but it is important to find out what accuracy actually is and how different errors have entered in to measurement.
- A study of errors is a first step in finding ways to reduce them.
- Errors may arise from different sources and are usually classified as under.



Gross Error:

- It mainly covers human mistakes in reading instruments and recording and calculating measurement results.

Gross errors can be avoided by using two suitable measures, and they are written below:

- Proper care should be taken in reading, recording the data. Also, the calculation of error should be done accurately.

- By increasing the number of experiments, we can reduce the gross error. If each experimenter takes different points, then by taking the average of more readings we can reduce the gross

Systematic Error:

Systematic errors can be better understood if we divide them into subgroups.

(i) **Instrumental Error:** These errors arise due to three main reasons:

(a) **Due to inherent shortcomings in the instruments:** These errors arise due to faulty construction and calibration of the measuring instruments. These errors may cause the instrument to read too low or too high.

(b) **Due to misuse of the instruments**

(c) **Due to the loading effects of instruments**

(ii) **Environmental Errors:** This type of error arises in the measurement due to the effect of the external conditions includes temperature, pressure, and humidity and can also include an external magnetic field.

(iii) **Observational Errors:** These are the errors that arise due to:

(a) An individual's bias, lack of proper setting of the apparatus, or an individual's carelessness in taking observations.

(b)The measurement errors also include wrong readings due to Parallax errors.

Random Error:

(a) The random errors are those errors, which occur irregularly and hence are random.

(b) The random errors are accidental, small and independent.

(c) They vary in an unpredictable manner.

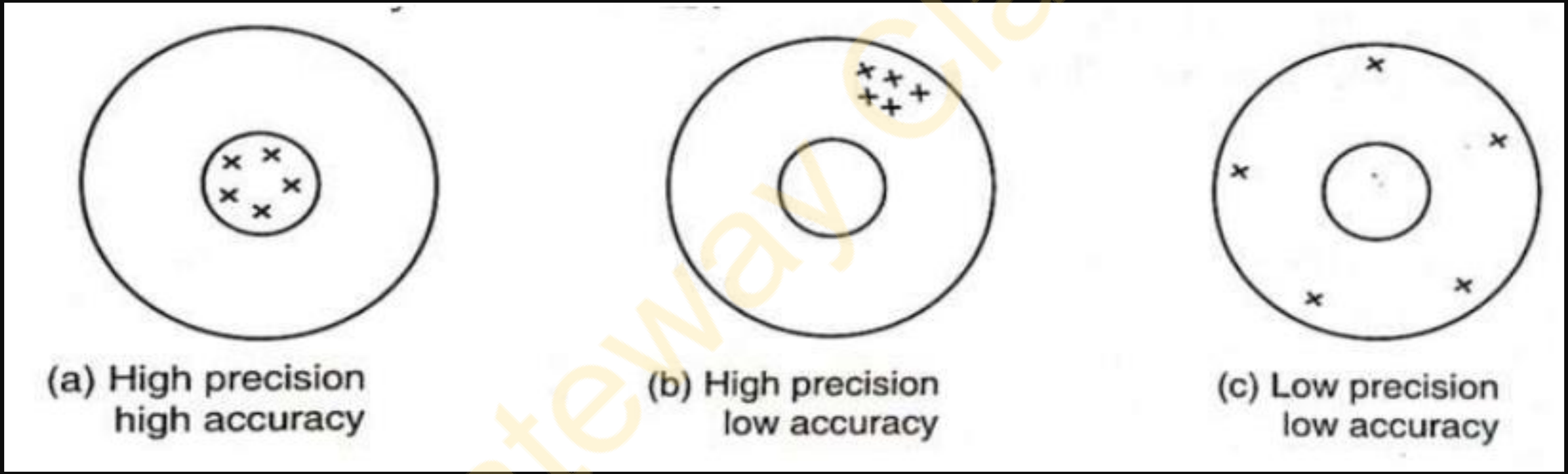
Differences between Systematic and Random Error

AKTU 2022-23 E

Basis for Comparison	Random Error	Systematic Error
Definition	Random error occurs in the experiment because of the uncertain changes in the environment	It is a constant error which remains same for all measurements
Causes	Environment, limitation of the instrument, etc.	Incorrect calibration and incorrectly using the apparatus
Minimize	By repeatedly taking the reading.	By improving the design of the apparatus
Magnitude of Error	vary	Constant
Direction of Error	occur in both the direction	occur only in one direction
Types	Do not have any type	Three

Accuracy and Precision

- In a set of measurements, accuracy is the closeness of the measurements to a true value.
- In a set of measurements precision is the closeness of the measurements to each other.



Calibration in Measurement

- Instrument calibration is one of the primary processes used to maintain instrument accuracy.

Or

- Calibration is the process of comparing the measurements of an instrument or device to a known standard or reference to determine any deviation from accuracy and to make necessary adjustments to correct these deviations.
- The goal of calibration is to ensure that the instrument provides accurate and reliable measurements within specified tolerances.

- Calibration provides consistency in readings and reduces errors, thus validating the measurement universally.

Measurement of Pressure

1. U-tube Manometer
2. Bourdon Tube Pressure Gauge

Manometer

- It is an instrument used for measuring the pressure at a point in a fluid by balancing the column of fluid by same or another fluid column.

- They are classified as

1. Simple Manometers

(a) Piezometers
(b) U-tube monomer
(c) Single column manometer

2. Differential manometers

(a) U-Tube differential manometer
(b) Inverted U-Tube differential manometer

U-Tube Manometer

➢This manometer consists of U- shaped glass-tube as shown in figure.

➢One end of the U-tube is connected to the point where pressure has to be measured while other end remains open to the atmospheric pressure.

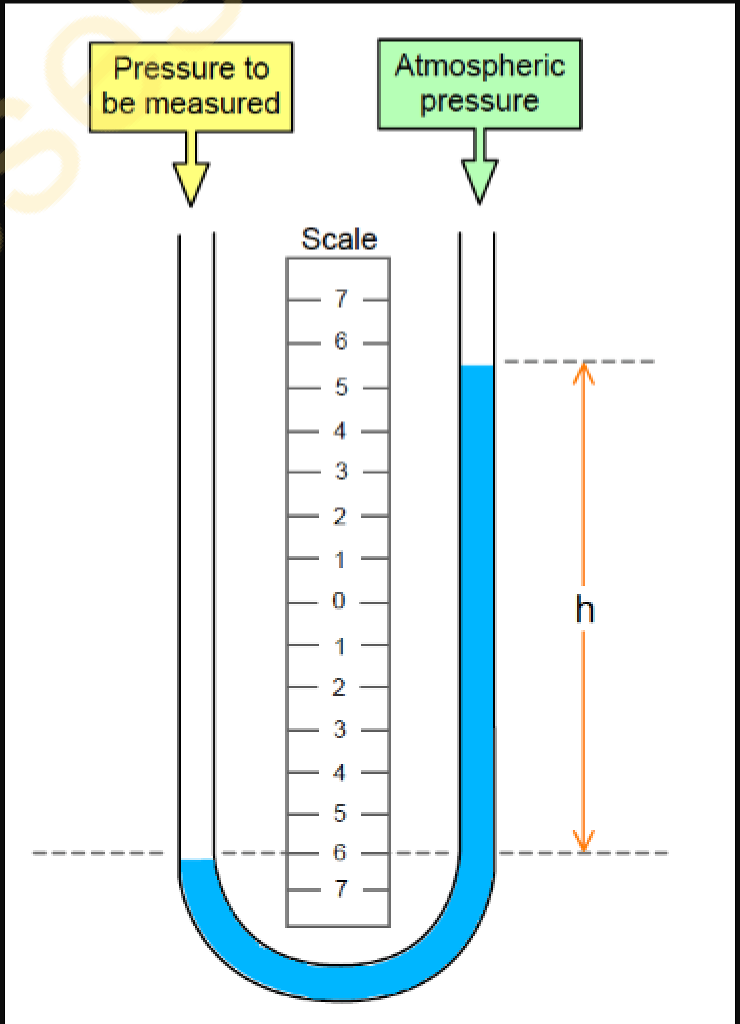
➢Water and mercury are used as a manometric fluid.

Advantages of Manometer

➢Simple in construction
➢High accuracy & sensitivity

Disadvantages of Manometer

➢Large & bulky
➢Measured fluids must be compatible with the manometer fluids
➢Need for leveling

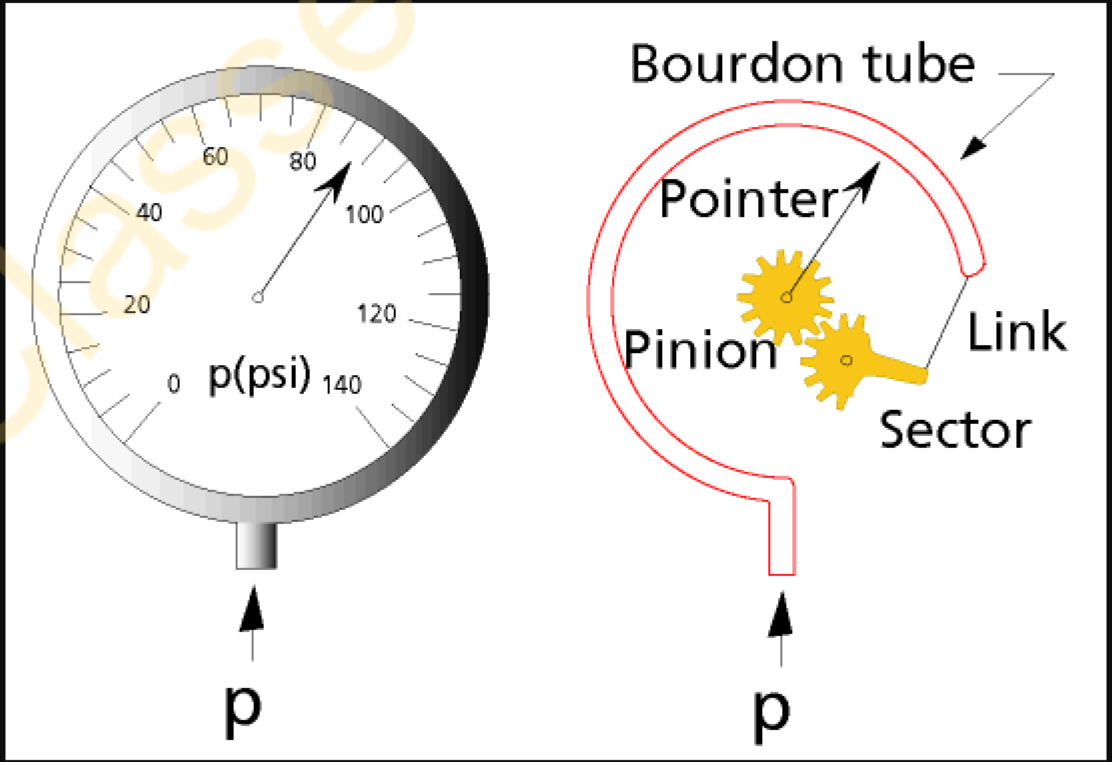


Bourdon Tube Pressure Gauge

- Bourdon tube pressure gauge is used to measure the pressure from 0.6 bar to 7000 bar.
- It is a mechanical pressure-measuring instrument.
- It operates without any electric power.

- When the bourdon tube is subjected to applied pressure then it deflects.
- This deflection is proportional to applied pressure.
- C-type bourdon tube consists of a long thin wall cylinder that is sealed at one end.
- The other end of the tube is fixed and open at the application where pressure is measured.

- The tip of the tube is connected to a segmental liver through an adjustable link.
- The segmental liver is provided with a rack that meshes with a suitable pinion.
- This pinion is mounted on a spindle.
- This spindle holds a pointer.
- The resulting movement of the free end of the tube causes the pointer to move over a scale



……..Bourdon Tube Pressure Gauge

Advantages of Bourdon Tube

- Low cost
- Simple in construction
- Wide rangeability
- Good accuracy
- Adaptable to transducer designs

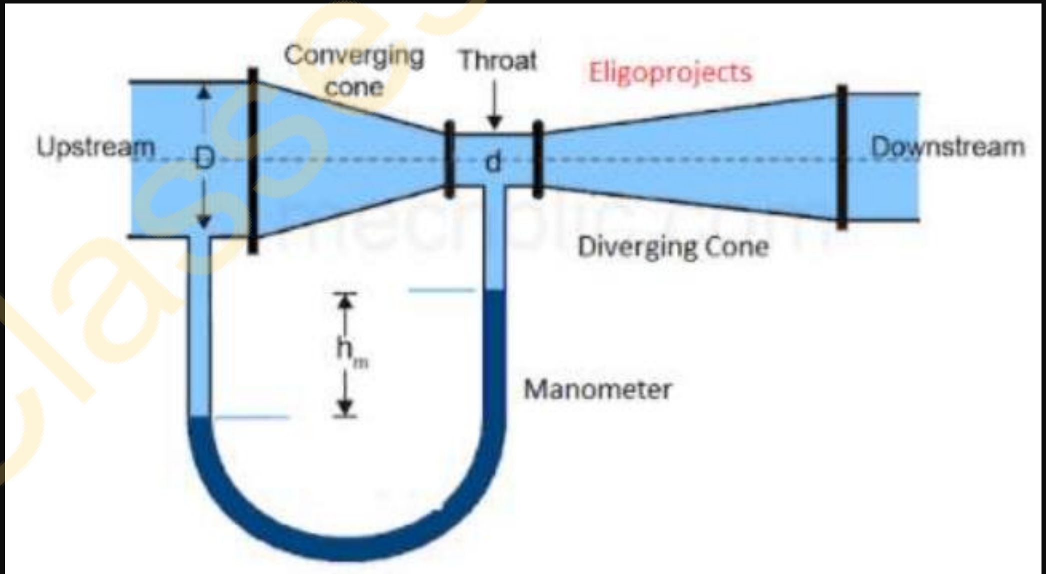
Disadvantages of Bourdon Tube

- Subject to Hysteresis
- Susceptible to shock & vibration

Measurement of mass flow rate : Venturi Meter

AKTU 2023-24 E

- It is a gradually converging-diverging device that is used to measure the discharge of fluid flow.
- Venturi meter is work on Bernoulli's equation.
- It is one of the costlier device to measure discharge.
- Accuracy is high.
- Losses are less.
- Value of C_d is high (0.94 to 0.98) [Coefficient of discharge ,C_d = Q_{act} / Q_{th}



$$Q = \frac{A_1 A_2 \sqrt{2gh}}{\sqrt{A^2{}_1 - A^2{}_2}}$$

Main parts of Venturimeter:-

1. Converging part
2. Throat
3. Diverging Part

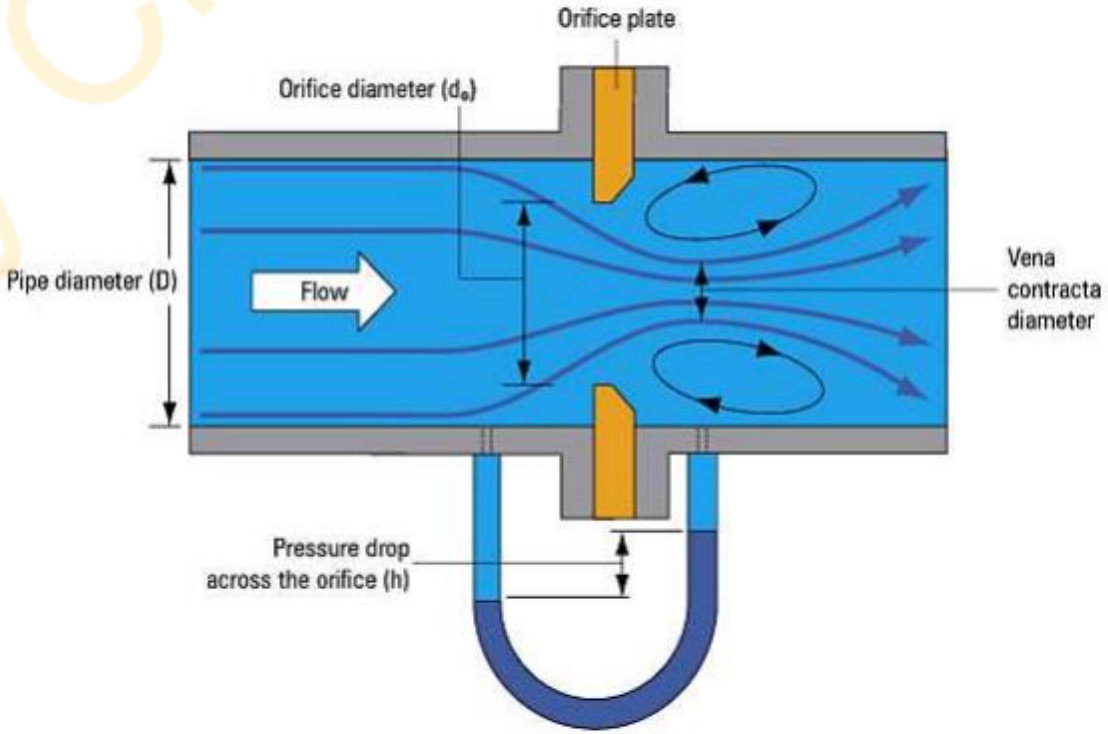
- Cross cross-sectional area of the throat section is smaller than the inlet section due to this the velocity of flow at the throat section is higher than the velocity at the inlet section, this happens according to the continuity equation.
- The increases in velocity at the throat result in decreases in pressure at this section, due to this pressure difference is developed between the inlet valve and throat of the venturi-meter.
- This pressure difference is measured by a manometer by placing this between the inlet section and the throat.
- Using the pressure difference value, we can easily calculate the flow rate through the pipe.

Measurement of mass flow rate : Orifice Meter

$$Q = C_c A_0 \frac{\sqrt{2gh}}{\sqrt{1 - C_c^2\left(\frac{A_0}{A_1}\right)^2}}$$

- Orifice Meter is a cheapest device to measure **discharge**.
- Losses are high.
- Value of C_d is low (0.60 to 0.65)   [Coefficient of discharge ,$C_d = \frac{Q_{act}}{Q_{th}}$]

- Orifice meter is device used to determine the rate of flow through pipe.
- It consist of flat circular plate which has a sharp edged circular hole called orifice.
- It is fixed concentric to pipe.
- The orifice diameter is generally kept half of the diameter of the pipe.
- It is based on the same principle as explained in venturimeter.
- The value of Cd varies between 0.60 to 0.65.
- It is a economical and less space is requred for fitting.



Measurement of Temperature: 1. Thermocouple Thermometer , 2. Optical Pyrometers

Temperature of a body is defined as degree of hotness or coldness of the body measured on a definite scale.

The commonly used scales for measuring temperature are:

- Kelvin scale (K)
- Centigrade scale (^{O}C)
- Fahrenheit scale (^{O}F)
- Rankine scale (R)

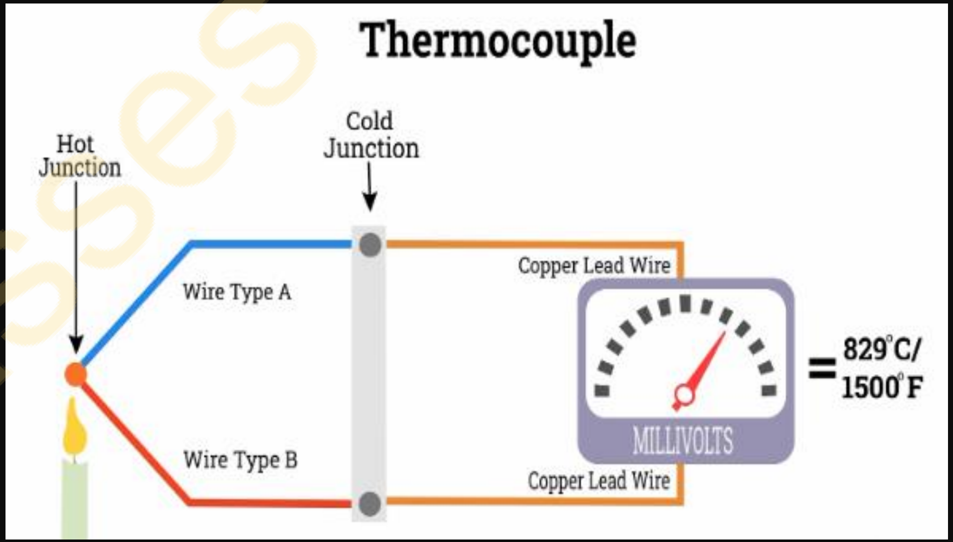
- The principle of temperature measurement is based on Zeroth's law of thermodynamics.

- The instrument which is used for measuring temperature is a thermometer.

Note: For a higher range of temperatures, i.e. above 650 ^{O}C, filled thermometers are unsuitable.

for the higher range of temperatures, thermocouples and pyrometers are used.

Thermocouple Thermometer



- A Thermocouple is a temperature measurement device that works on the principle of the Seebeck Effect.
- It is widely used in industrial and scientific applications due to its ability to measure a wide range of temperatures accurately.

Working Principle

Seebeck Effect: When two different metal wires (Type A and Type B) are joined at one end and exposed to heat, an electromotive force (e.m.f.) is generated.

This phenomenon of generation of e.m.f. is called the Seebeck effect.

Voltage Measurement: The voltage produced is proportional to the temperature difference between the hot and cold junctions.

Temperature Reading: The generated voltage is measured using a milli-voltmeter, which is calibrated to display temperature readings.

Applications of Thermocouple

- ➢ Used in furnaces, boilers, and high-temperature equipment.
- ➢ Used in laboratories and meteorology for temperature measurement.
- ➢ Measures temperature in engines and exhaust gases.

Advantages of Thermocouple

✓ Can measure temperatures from -200°C to over 2000°C.

✓ Detects temperature changes instantly.

✓ No external power source required.

Limitations of Thermocouple

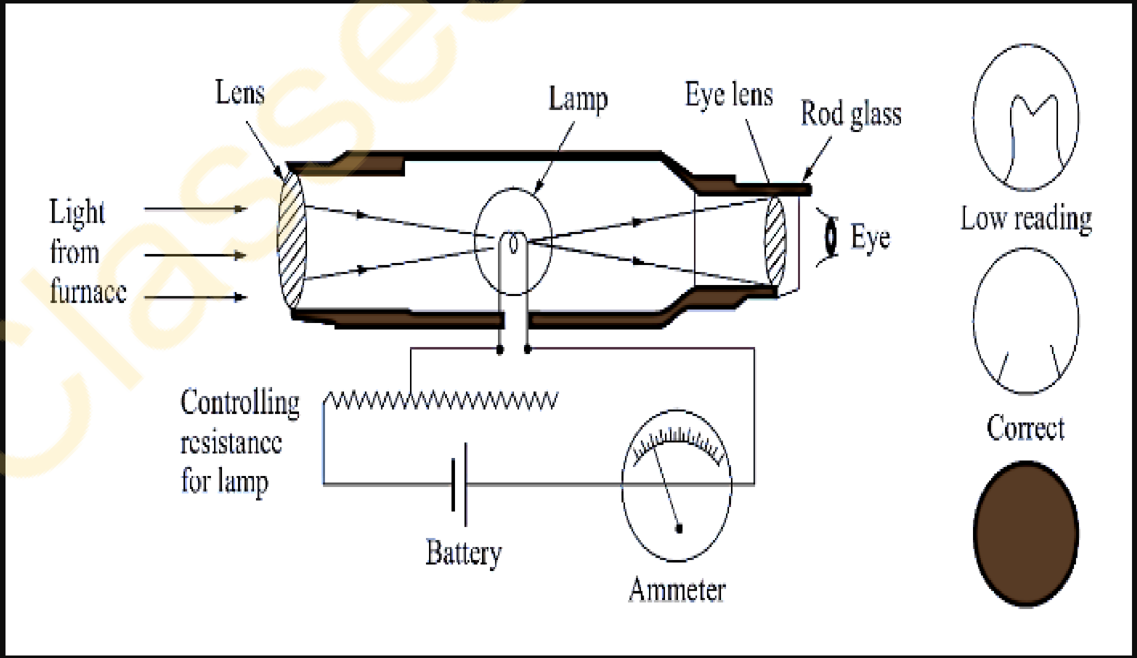
✗ Can be affected by nearby electromagnetic waves.

✗ Requires additional correction for precise measurement.

Optical Pyrometers

AKTU 2023-24 Odd

Construction of Optical Pyrometers :



Lens

- Collects light from the hot object.

Lamp & Filament:

- Adjusts brightness for temperature measurement.

Controlling Resistance

- Controls the filament brightness.

Battery

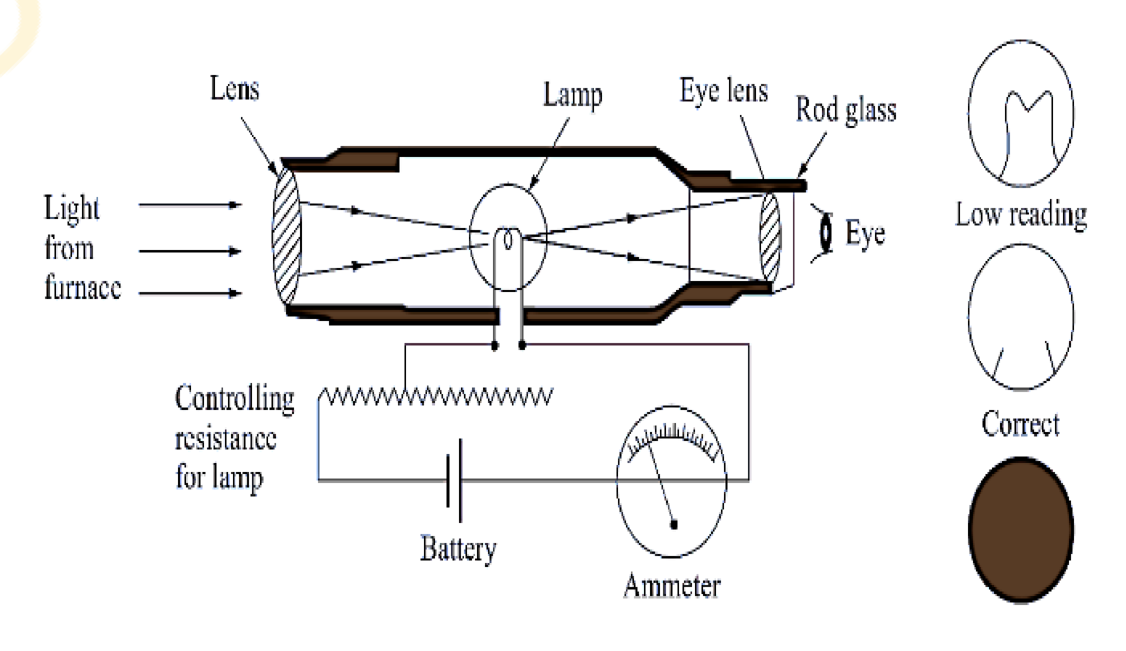
- Supplies power to the electrical components.

Ammeter

- Measures the electric current and displays the temperature.

Working Principle of Optical Pyrometers :

- If we can measure brightness of the light of a given emitted by a hot source , we can estimate its temperature.
- This is the principle of optical pyrometers.
- It can be used to measure temperature in the range of 800 ^oC to 1200 ^oC.
- Light from the hot object (e.g., furnace) is collected through a lens and directed into the pyrometer.
- A tungsten filament inside the pyrometer is heated by adjusting a controlled resistance.
- The operator adjusts the filament temperature until its brightness matches that of the hot object.
- When the brightness matches, the temperature is read from the calibrated scale.



Optical Pyrometers

Uses :

It is widely used for accurate measurement of temperature.

- Furnace
- Molten metal
- Steel manufacturing
- Glass production
- Ceramics manufacturing
- Furnace Monitoring
- Aerospace engineering

Advantages :

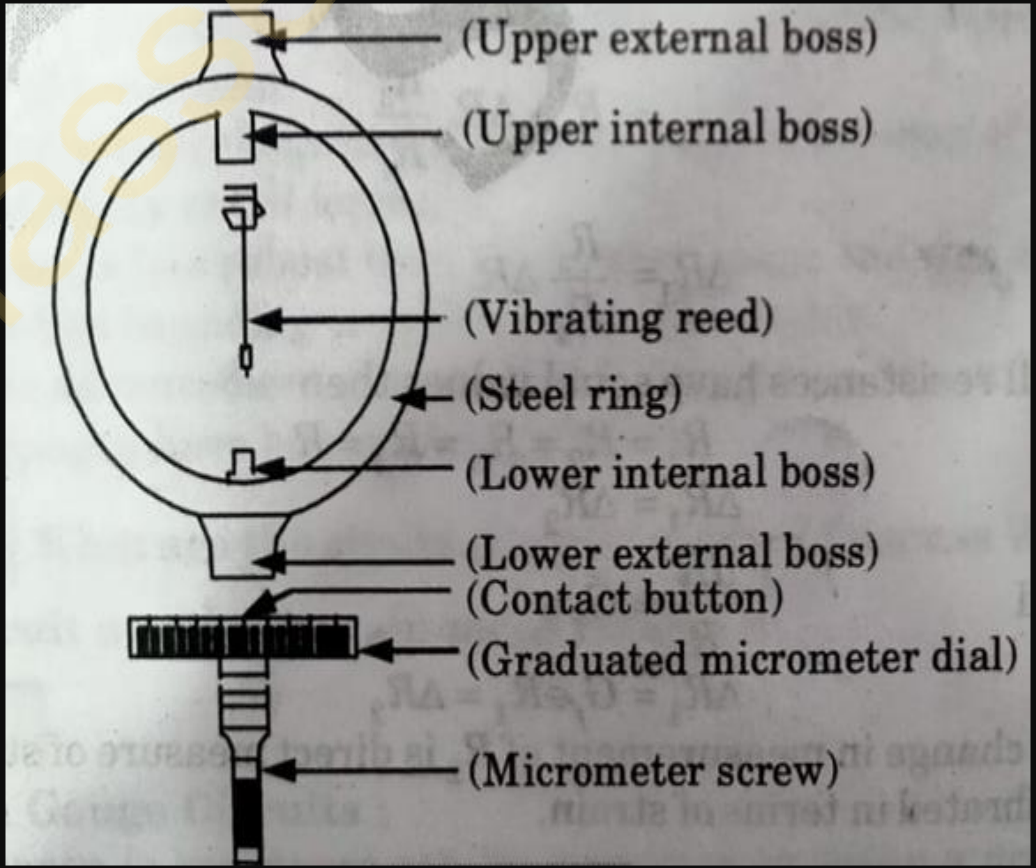
- Excellent accuracy within $\pm 5\ ^{o}C$
- No direct contact with the object whose temperature is to be measured.
- Measurement is independent of the distance between the body and the instrument.
- The semiskilled operator is required

Dis-Advantages :

- Can not measure lower temperature
- Radiation is harmful

Measurement of Force (Proving Ring)

- The proving ring is a device used to measure force.
- It consists of an elastic ring of known diameter with a measuring device located in the center of the ring.
- Proving rings come in a variety of sizes. They are made of a steel alloy.
- Proving rings can be designed to measure either compression or tension forces. Some are designed to measure both.



Construction of Proving Ring:

- The proving ring consists of two main elements, the ring itself and the diameter-measuring system.
- Forces are applied to the ring through the external bosses.
- The resulting change in diameter, referred to as the deflection of the ring, is measured with a micrometer screw and the vibrating reed mounted diametrically within the ring.
- The micrometer screw and the vibrating reed are attached to the internal bosses of the ring.

……Measurement of Force (Proving Ring)

Working of Proving Ring:

- To read the diameter of the ring, the vibrating reed is set in motion by gently tapping it with a pencil.
- As the reed is vibrating, the micrometer screw on the spindle is adjusted until the contact-button on the spindle just touches the vibrating reed, dampening out its vibrations.
- When this occurs a characteristic buzzing sound is produced.
- At this point a reading of the micrometer dial indicates the diameter of the ring.

Applications:

Proving rings are commonly used in various engineering and scientific fields for force measurement, including:

- Material testing
- Structural analysis
- Load monitoring in machinery and equipment
- Calibration of other force-measuring devices
- Geotechnical testing, such as soil and rock mechanics

Advantages:

- High accuracy and sensitivity
- A wide range of measurement capacities is available
- Suitable for both static and dynamic force measurements
- Robust construction for reliable performance in harsh environments

Measurement of Torques (Prony Brake Dynamometer)

Working and construction of Prony Brake Dynamometer:

- A Prony Brake Dynamometer is a simple mechanical device used to measure the torque and power output of rotating shafts, especially in engines and machines.

- It consists of two wooden blocks, placed around a pulley of radius R as shown in fig.
- The pulley is keyed to the shaft of the engine whose power to be measured.
- The wooden blocks are clamped by two nuts and bolts.
- A helical spring is placed between the nut and the upper wooden block in order to adjust the pressure on the pulley and hence control its speed.
- The upper block also has a long lever attached to it which carries a load ‘L’ at its outer end.
- A counter weight is placed on the other end of lever to balance the brake when unloaded.
- The stops limit the motion of the lever.
- To measure shaft power using prony brake dynamometer, the lever is loaded with suitable load ‘L’ and nuts tightened such that the engine shaft runs at a constant speed and the lever is in a horizontal position.

Measurement of Torques (Prony Brake Dynamometer)

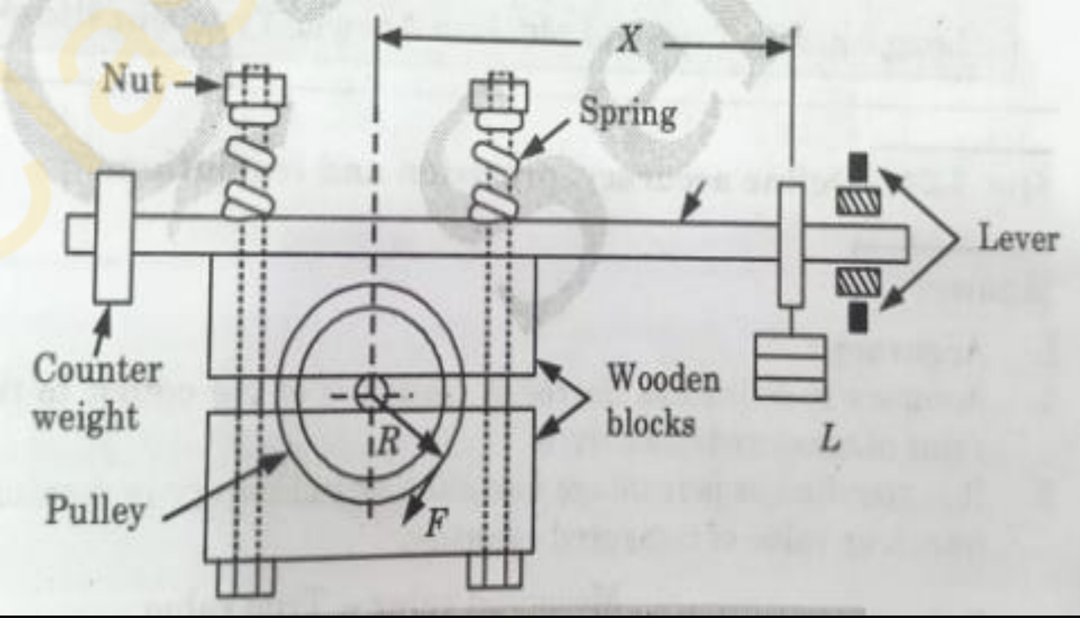
- Under such working conditions, the moment due to load ‘L' balances the moment due to frictional resistances force between wooden blocks and the pulley.

- Hence, torque(T) on the shaft is given by the relation
  $T = L \times X$ = F x R

- Brake Power(P) transmitted by the engine shaft,

$$P = \frac{2\pi NT}{60} \text{ watt.}$$

N = rpm.



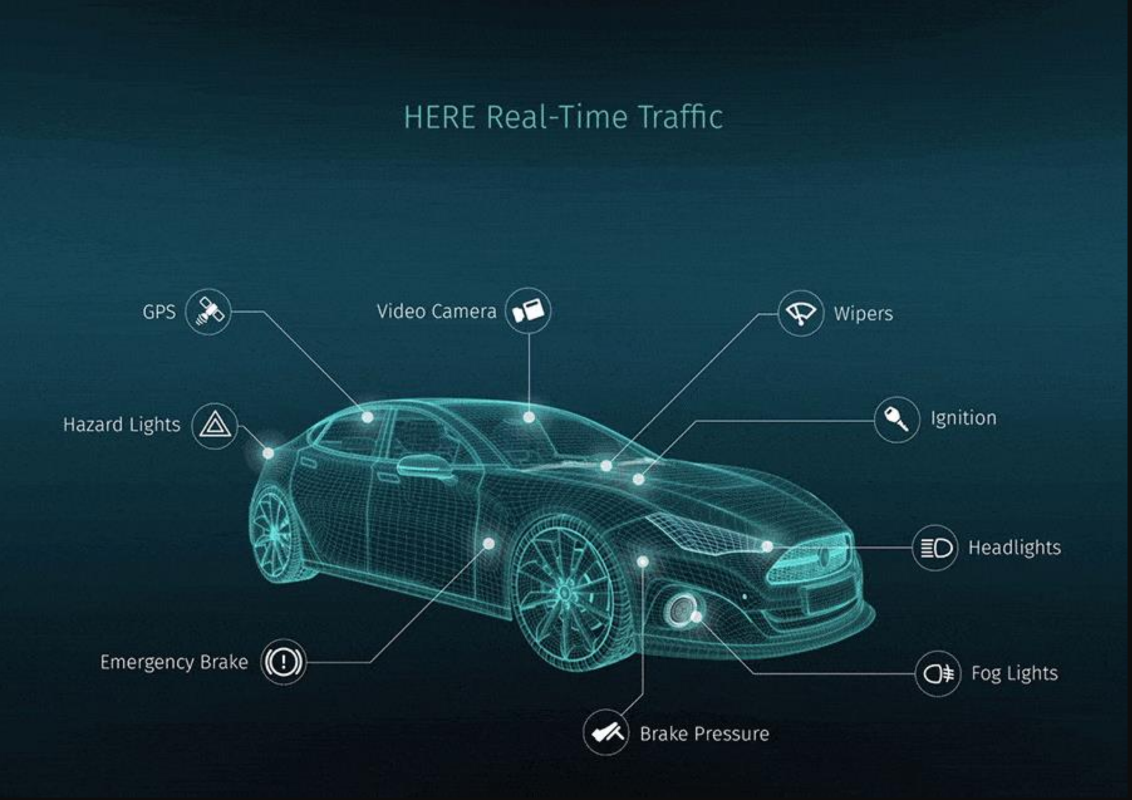
Unit-5: Introduction to Measurement and Mechatronics

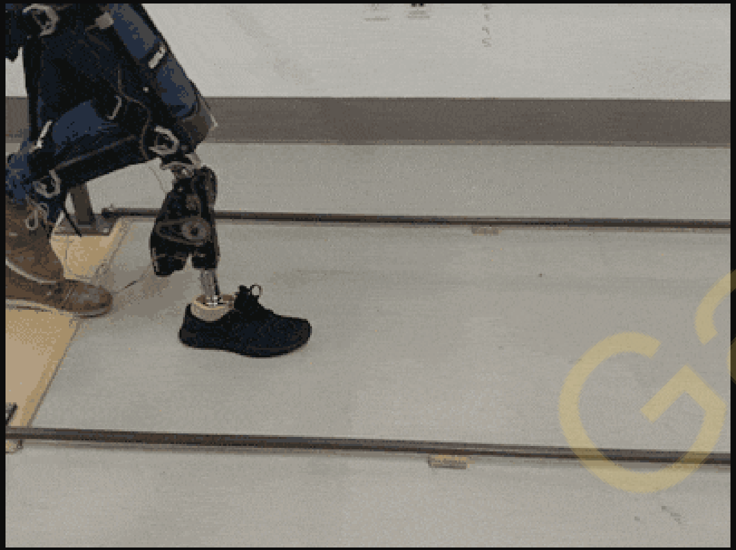
Introduction to Measurement: Concept of Measurement, Error in measurements, Calibration, measurements of pressure(Bourdon Tube Pressure and U-Tube Manometer), temperature(Thermocouple and Optical Pyrometer), mass flow rate(Venturi Meter and Orifice Meter), strain(Bonded and Unbonded Strain Gauge), force (Proving Ring) and torques(Prony Brake Dynamometer); Concepts of accuracy, precision and resolution.

Introduction to Mechatronic Systems: Evolution, Scope, Advantages and disadvantages of Mechatronics, Industrial applications of Mechatronics, Introduction to autotronics, bionics, and avionics and their applications. Sensors and Transducers: Types of sensors, types of transducers and their characteristics.

Overview of Mechanical Actuation System – Kinematic Chains, Cam, Ratchet Mechanism, Gears and its type, Belt, Bearing.

Hydraulic and Pneumatic Actuation Systems: Overview: Pressure Control Valves, Direction Control Valves, Rotary Actuators, Accumulators and Pneumatic Sequencing Problems.


HERE Real-Time Traffic
GPS
Video Camera
Wipers
Hazard Lights
Ignition
Headlights
Emergency Brake
Fog Lights
Brake Pressure

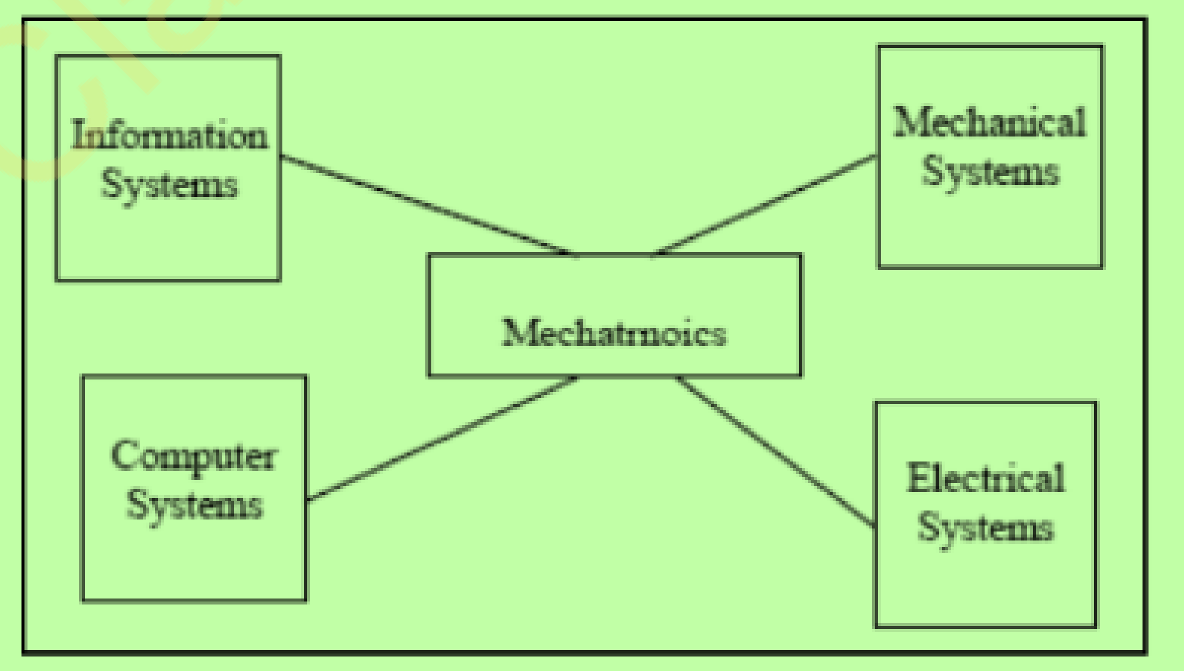


Q.1 Define mechatronics. Also write down the advantages, disadvantages, and applications of mechatronics.

AKTU-2020-21, 2021-22, 2022-23 7Marks

Mechanics + electronics = Mechatronics

- The term "mechatronics" itself is a blend of "mechanics" and "electronics.



- **Mechatronics is an interdisciplinary field that combines mechanical engineering, electronics, computer science, and control systems to create smart and automated systems.**
- **Concurrently it includes all these disciplines.**

Examples of mechatronics

1. In modern automobiles, mechanical fuel injection systems are now replaced with electronic fuel injection systems.

➢ This replacement made the automobiles more efficient and less pollutant.

2. Industrial Robots: Automated assembly and welding in manufacturing industries using by Sensors, Microcontroller and Actuators.

3.Today's domestic washing machines are "intelligent"

4. Today's four-wheel passenger automobiles are equipped with safety installations such as airbags, parking (proximity) sensors, antitheft electronic keys, etc.

5.ATM Machines: Automating banking transactions.

6.Self-Driving Cars: Autonomous vehicle control.

Advantages of Mechatronics

1. **Integrated Design:** Mechatronic systems integrate mechanical, electrical, and computing components, leading to more efficient and compact designs.

2. **Enhanced Functionality:** Mechatronic systems can perform complex tasks with greater precision and flexibility compared to purely mechanical systems.

3. **Automation:** Mechatronic systems are often used in automation processes, leading to increased productivity, reduced labor costs, and improved safety.

4. **Improved Performance:** The integration of sensors and control systems in mechatronic designs allows for real-time monitoring and adjustment, leading to improved performance and reliability.

5. **Adaptability:** Mechatronic systems can be easily reprogrammed or reconfigured to adapt to changing requirements.

6. It is also more user-friendly and safer to use.

Disadvantages of Mechatronics

1.Complexity: Designing and integrating multiple disciplines can increase the complexity of mechatronic systems, leading to challenges in development, maintenance, and troubleshooting.

2.Cost: High initial cost -Mechatronic systems can be more expensive to develop and manufacture compared to purely mechanical systems due to the need for additional components such as sensors, actuators, and control systems.

3.Skills Requirement: Developing mechatronic systems requires expertise in multiple disciplines, which may be challenging to find in one individual or team.

4. It is expensive to incorporate a mechatronic approach into an existing/old system.

5. Specific problems for various systems would have to be addressed separately and properly.

Industrial Applications of Mechatronics

1.Manufacturing Automation: Mechatronic systems are widely used in manufacturing processes for automation, precision control, and quality assurance. Industrial robots equipped with mechatronic systems perform tasks such as welding, assembly, painting, and material handling in factories, leading to increased productivity and reduced labor costs.

2.CNC Machines: Computer Numerical Control (CNC) machines, including CNC milling machines, lathes, and machining centers, utilize mechatronic systems for precise control of cutting tools and work pieces. Mechatronics enables CNC machines to execute complex machining operations with high accuracy and repeatability.

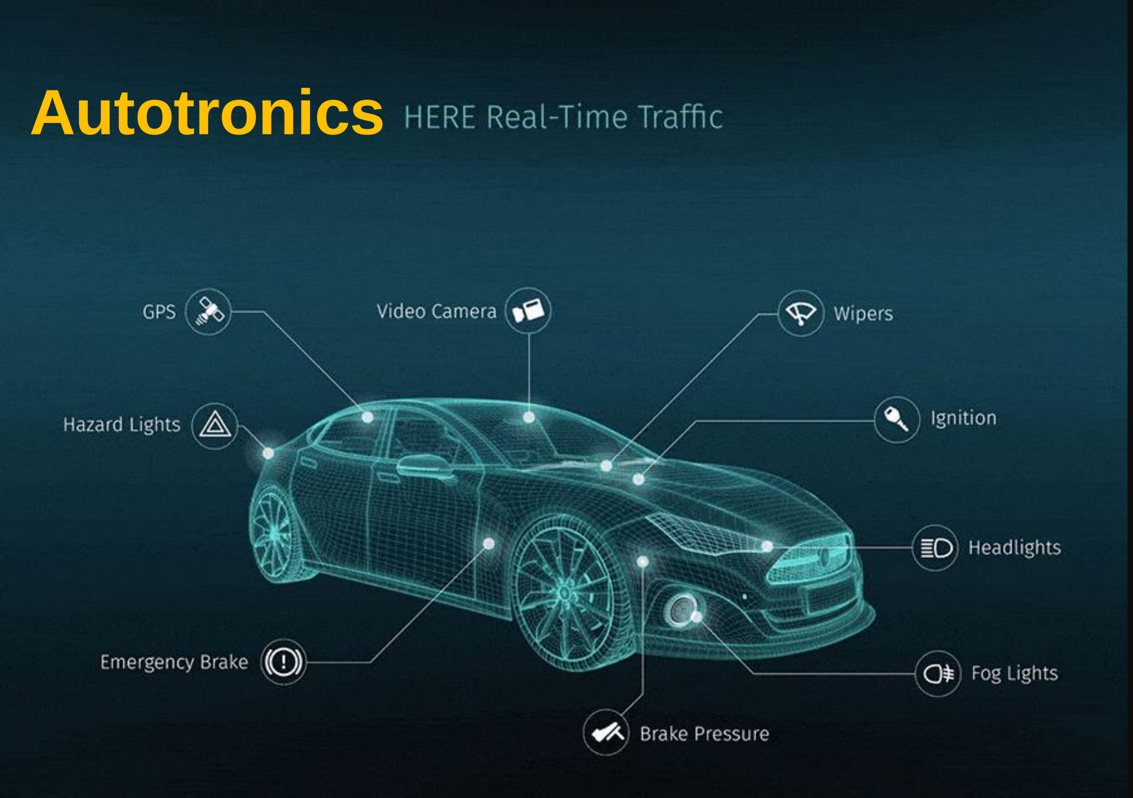
3.Process Control: Mechatronics plays a crucial role in industrial process control systems, including chemical plants, refineries, and power plants. Mechatronic sensors, actuators, and control algorithms monitor and regulate parameters such as temperature, pressure, flow rate, and chemical composition to optimize production processes and ensure safety and efficiency.

4. Packaging Machinery: Mechatronic systems are integrated into packaging machinery to automate packaging processes in industries such as food and beverage, pharmaceuticals, and consumer goods.

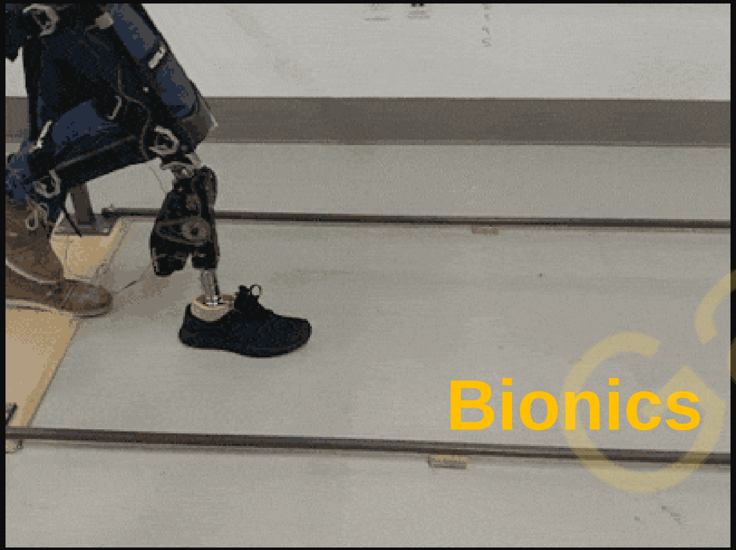
5. Material Handling Systems: Mechatronics is used in material handling systems such as conveyor belts, and automated guided vehicles (AGVs in warehouses, distribution centers, and production facilities. Mechatronic material handling systems enhance efficiency, reduce errors, and optimize logistics operations.

6. 3D Printing/Additive Manufacturing: Mechatronic systems are utilized in 3D printing and additive manufacturing technologies to control the deposition of materials layer by layer, Mechatronics plays a vital role in ensuring the accuracy, speed, and reliability of 3D printing processes across various industries.

7. Automated Inspection and Quality Control: Mechatronic systems are integrated into inspection and quality control systems to detect defects, measure dimensions, and ensure product conformity in manufacturing processes.


Autotronics
HERE Real-Time Traffic
GPS
Video Camera
Wipers
Hazard Lights
Ignition
Headlights
Emergency Brake
Fog Lights
Brake Pressure


Avionics


Bionics

Q.2 Write short notes on (i) Autotronics (ii) Bionics (iii)Avionics along with their applications.

AKTU-2020-21, 2021-22, 2022-23 7Marks

(i) Autotronics

Autotronics is the integration of automotive technology with electronics. It involves the use of electronic systems in vehicles for various functions such as engine management, transmission control, and safety features like airbags and ABS.

Applications: **Enhancing vehicle performance, improving fuel efficiency, increasing safety features, and enabling advanced functions like autonomous driving.**

(ii) Bionics

Bionics is a field that merges biology with engineering and technology to create systems, devices, and materials that interface with biological systems.

Applications: Bionics aids in addressing body damage by developing highly functional prosthetic limbs, and restoring mobility and quality of life for individuals with limb loss. Additionally, bionics contributes to tissue regeneration through the development of bio-inspired materials and technologies, offering promising solutions for repairing damaged tissues and organs.

(iii) Avionics

Avionics refers to the electronic systems used in aircraft for communication, navigation, monitoring, and control.

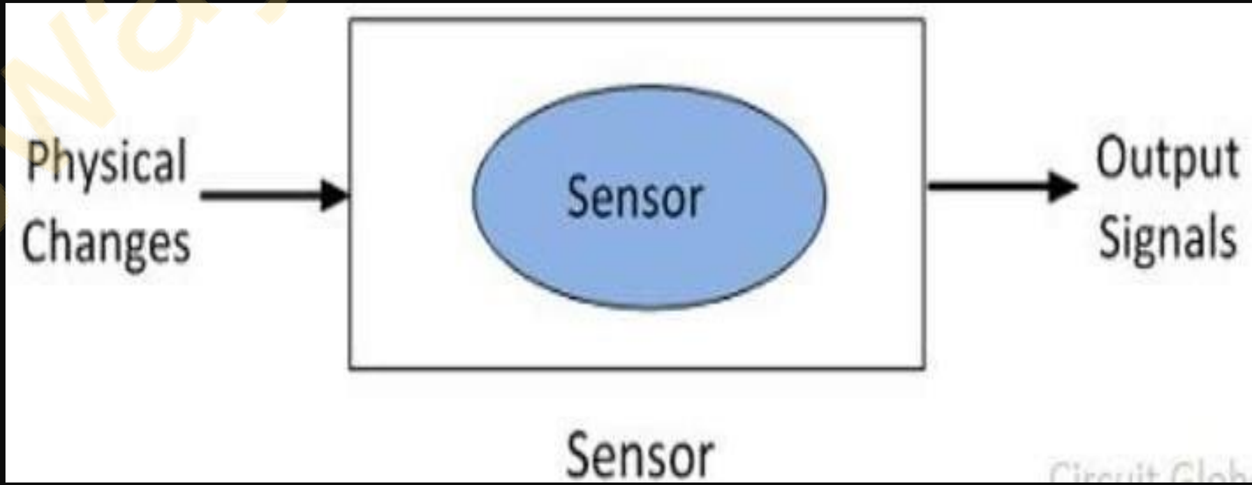
Applications: Essential for safe air travel. Used in commercial planes, military aircraft, drones, and even space exploration for navigation, communication, and flight control.

Q.3 What are Sensors and Transducers and their types? Also, explain the characteristics of sensors and transducers.

AKTU-2022-23 Odd

Sensor: A sensor is a device that provides usable output in response to change in a specified physical quantity which is measured. A device that receives and responds to a signal.

- The physical quantity may be temperature, force, pressure, displacement, flow, etc.
- For example, the **bulb of a mercury thermometer** senses the temperature of the body
  in contact.

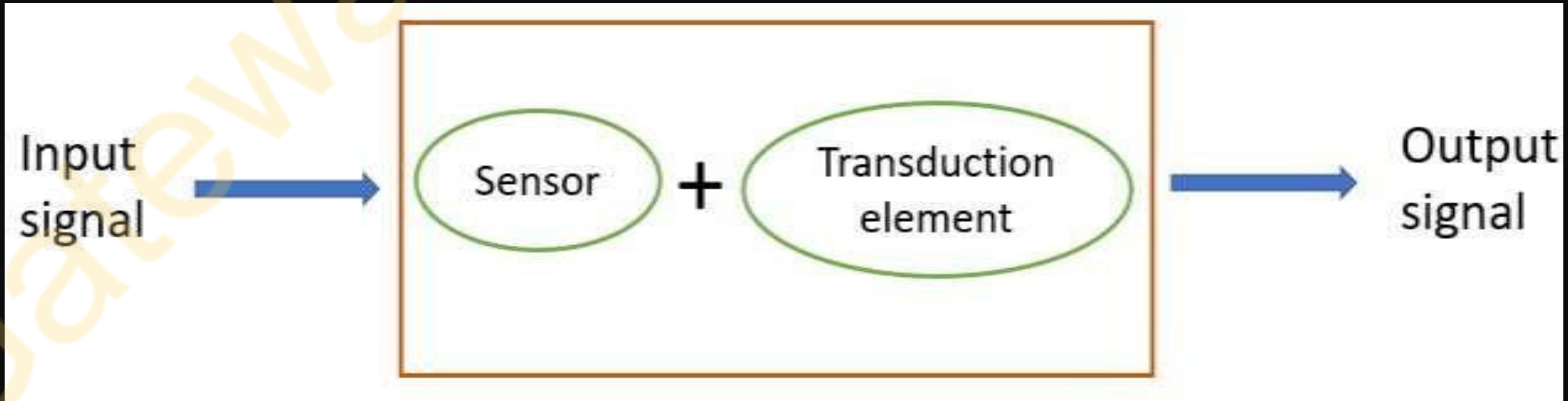


- The transducer is a device that changes the physical attributes of the non-electrical signal into an electrical signal which is easily measurable.
- Transducers convert one form of energy into another
- The process of energy conversion in the transducer is known as the transduction(figure).

It consists of two parts:

1. Sensing element/detector
2. Transduction element

For example : Thermocouple Thermometer



Types of Sensors

1. **Temperature Sensors:** Measure temperature changes.
2. **Pressure Sensors:** Measure pressure variations.
3. **Proximity Sensors:** Detect the presence or absence of an object without physical contact.
4. **Motion Sensors:** Detect motion or movement.
5. **Light Sensors:** Measure light intensity or detect the presence of light.
6. **Humidity Sensors:** Measure relative humidity in the air.

AKTU 2023-24 Odd

Types of Transducers

1. **Piezoelectric Transducers:** Convert mechanical stress or pressure into electrical energy and vice versa.
2. **Thermoelectric Transducers:** Convert temperature differences into electrical voltage.
3. **Strain Gauge Transducers:** Measure strain or deformation in an object.
4. **Optical Transducers:** Convert light energy into electrical signals and vice versa.

Capacitive Transducers: Measure changes in capacitance due to variations in physical quantities like displacement, pressure, or humidity.

5. **Inductive Transducers:** Measure changes in inductance due to variations in physical quantities like position or displacement.

Transducer based on need of an External Power Source

AKTU 2022-23

Active Transducer: Active transducers are those which do not require any power source for their operation. For example, a thermocouple, thermometer etc.

Passive Transducer: Transducers which require an external power source for their operation is called as a passive transducer. For example, a strain gauge, thermistor etc.

Characteristics of sensors and transducers

Characteristics are mainly divided into two categories:

i) Static characteristics **ii) Dynamic characteristics**

i. Static characteristics:

Static characteristics refer to the characteristics of the system when the input is either held constant or varying very slowly. For example Range, sensitivity, linearity, etc.

ii. Dynamic characteristics:

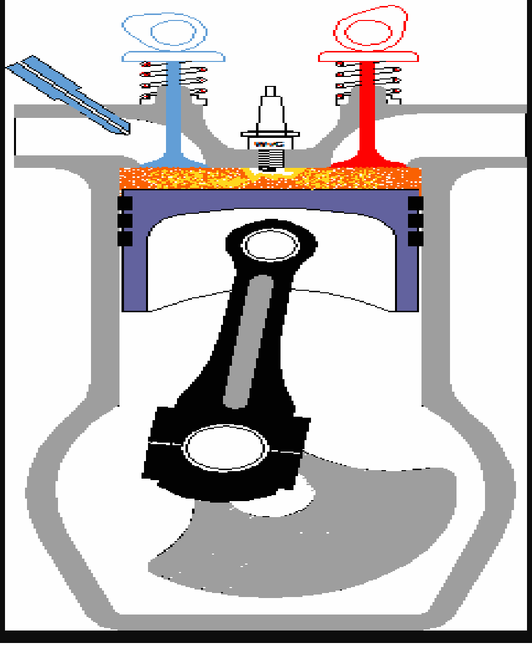
Dynamic characteristics describe the behavior of sensors and transducers in response to changes or variations in the input quantity over time. These characteristics are important for applications involving dynamic or time-varying signals.

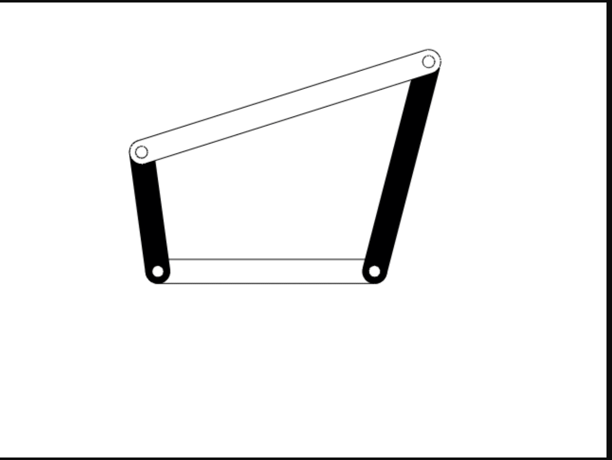
For example Response Time, Bandwidth, Frequency Response, etc.

Q.4 Write short notes on (i) Kinematic Chains (ii) Cam (iii) Ratchet Mechanism (iv)Gears and their type, (v)Belt (vi) Bearing.

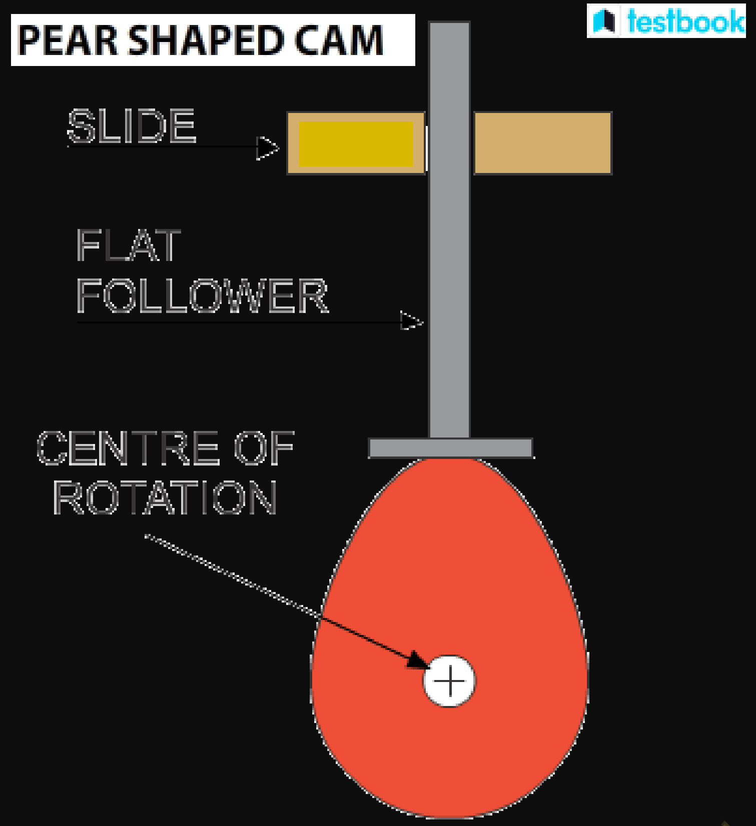
Kinematic Chain:

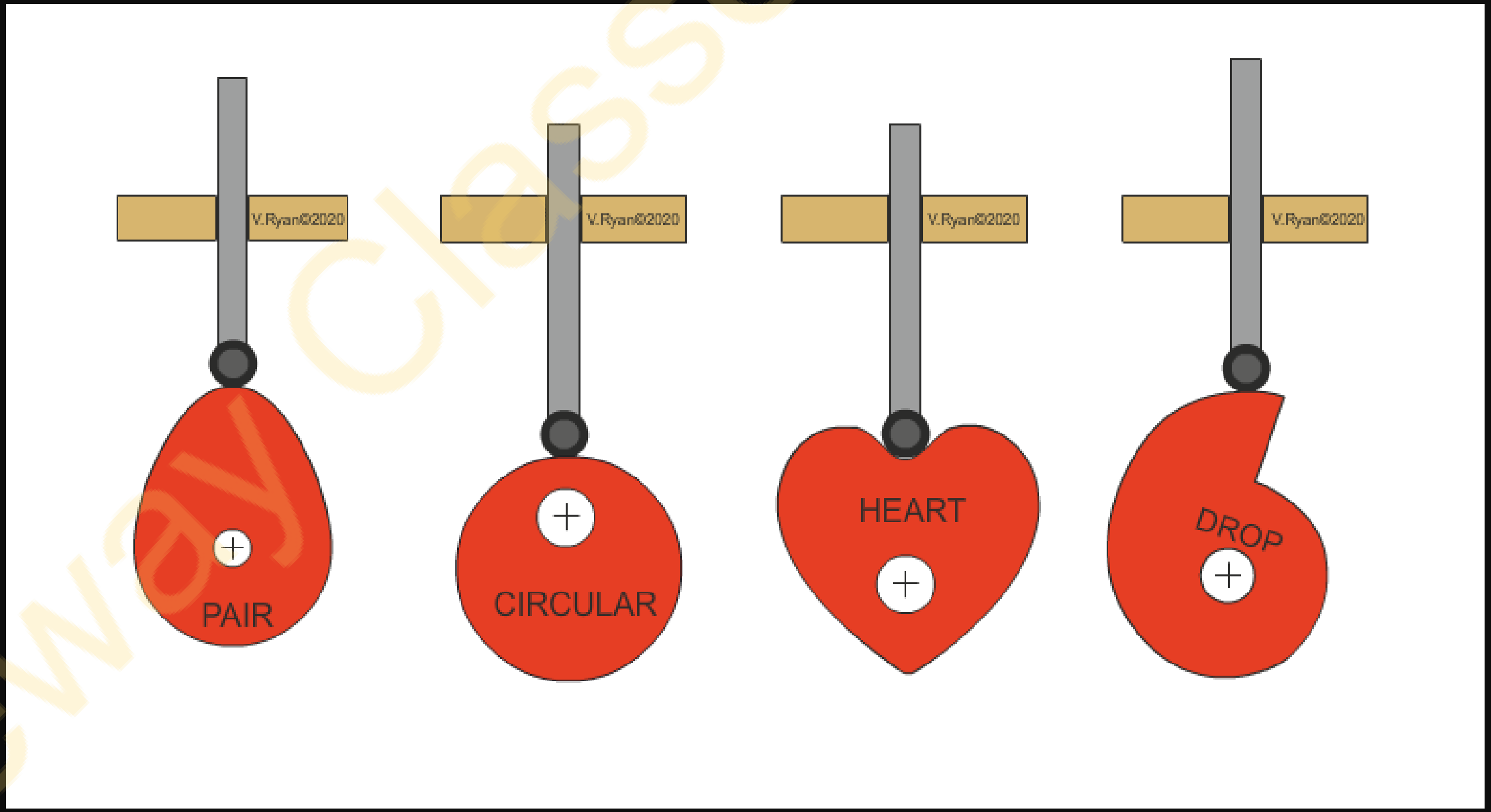
"If all the links are connected in such a way that first link is connected to last link in order to get the close chain and if all the relative motion in this close chain are constrained then such a chain is known as kinematic chain".





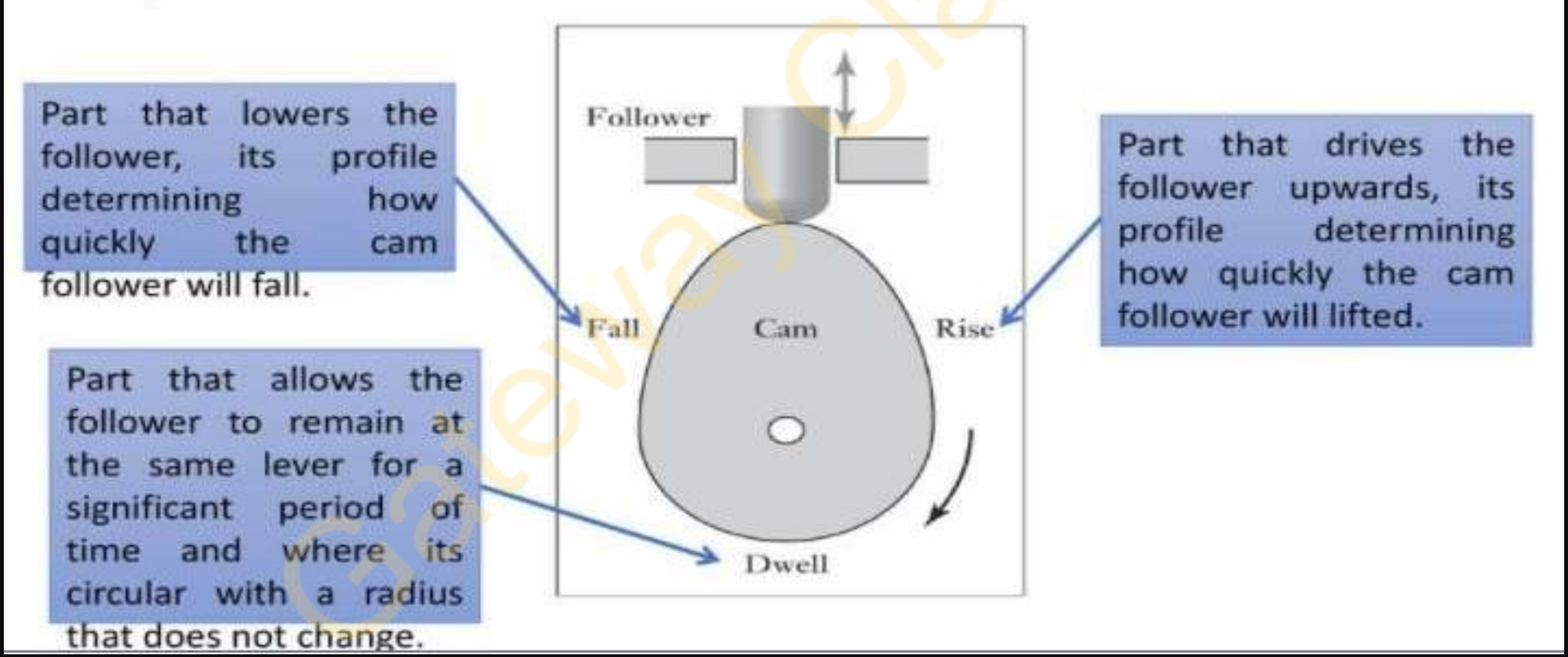
(ii) Cam


PEAR SHAPED CAM
SLIDE
FLAT FOLLOWER
CENTRE OF ROTATION


V.Ryan©2020
PAIR
V.Ryan©2020
CIRCULAR
V.Ryan©2020
HEART
V.Ryan©2020
DROP

(ii) Cam

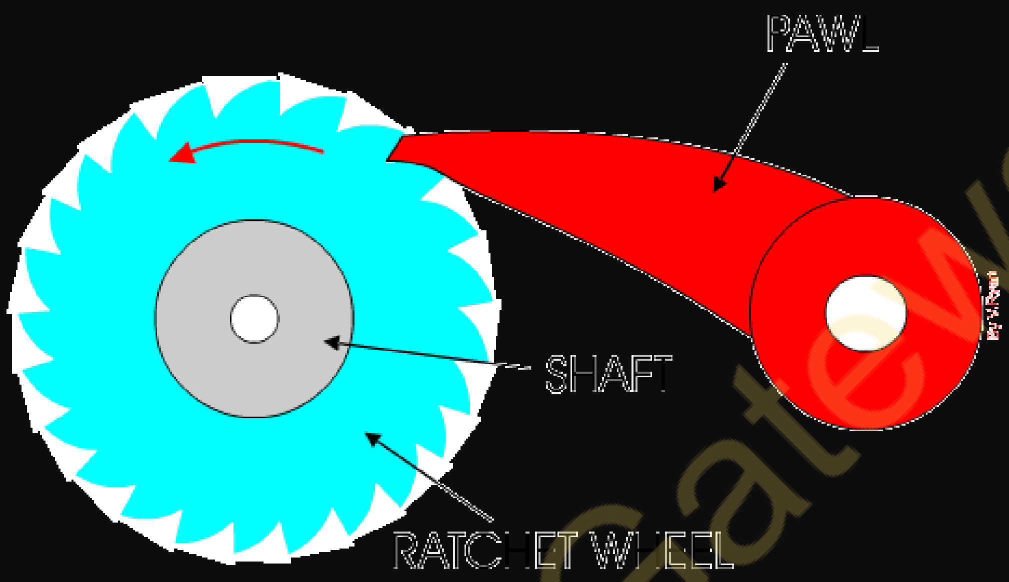
A cam is a mechanical device used to convert rotary motion into reciprocating motion or vice versa. It typically consists of a cylindrical or irregularly shaped component (the cam) mounted on a shaft, along with a follower that moves in contact with the surface of the cam. As the cam rotates, the shape of its surface causes the follower to move in a specific pattern, producing linear motion.

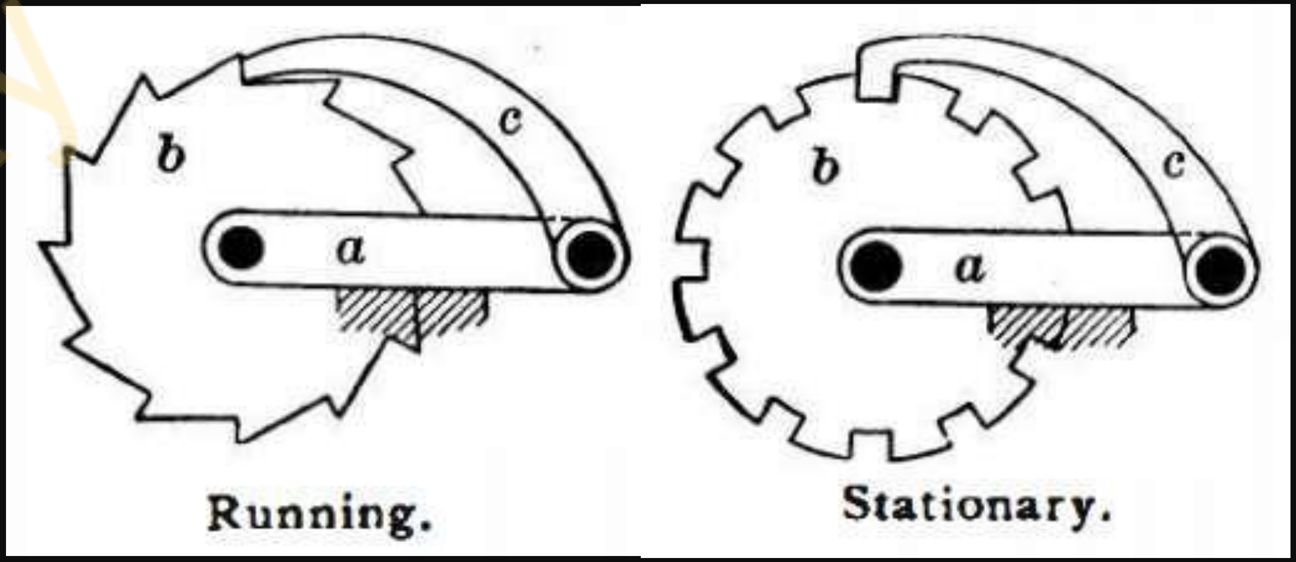


(iii) Ratchet and Powel Mechanism

AKTU 2023-24 E

- A ratchet mechanism is a mechanical device that allows motion in one direction while preventing motion in the opposite direction.
- It typically consists of a toothed wheel (the ratchet) and a pawl that engages with the teeth of the wheel.
- When the pawl engages with the teeth, it allows the wheel to rotate in one direction, but it prevents it from rotating in the opposite direction.





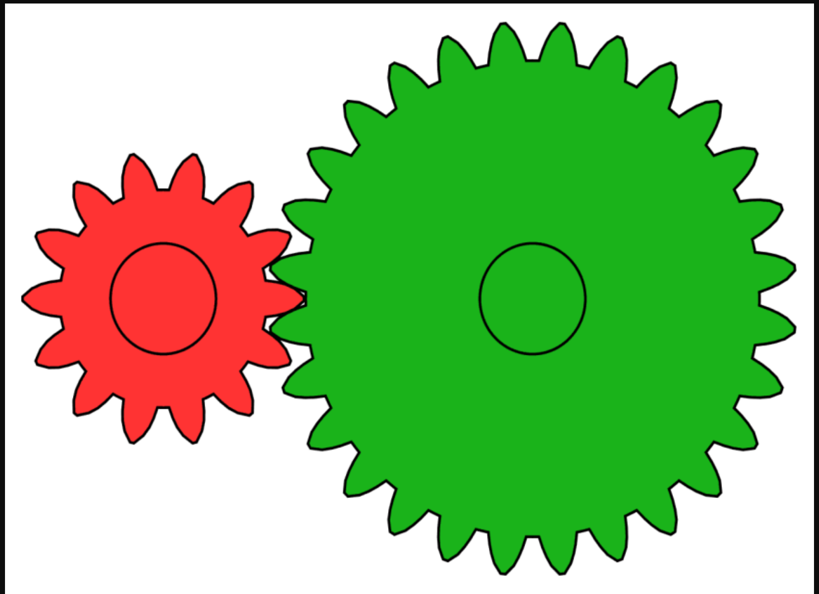
3
4

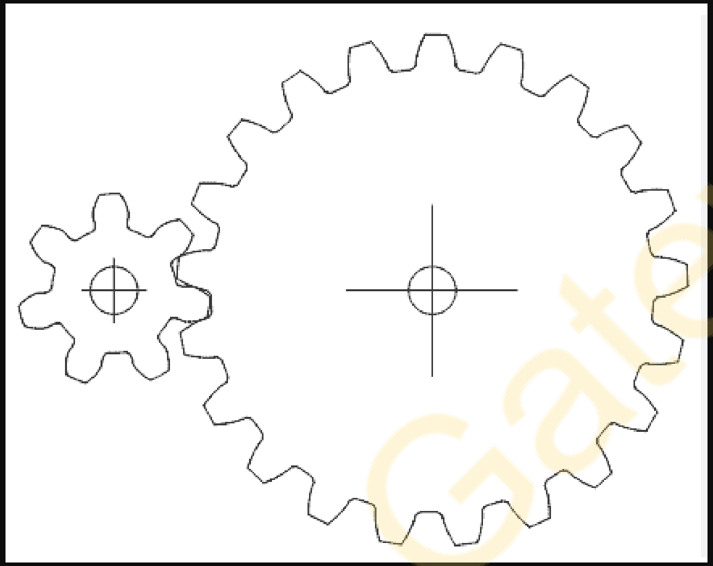
(iv) Gears and their type

AKTU-2022-23(E) 2Marks

Gears are mechanical components **used to**

- Transmit power
- Change the velocity
- Change the direction





Types of Gears

Types of Gears

Spur gear : This is Cylindrical gear. Teeth are parallel to axis. This is a highly demanded gear, which is easy to manufacture and to assemble.

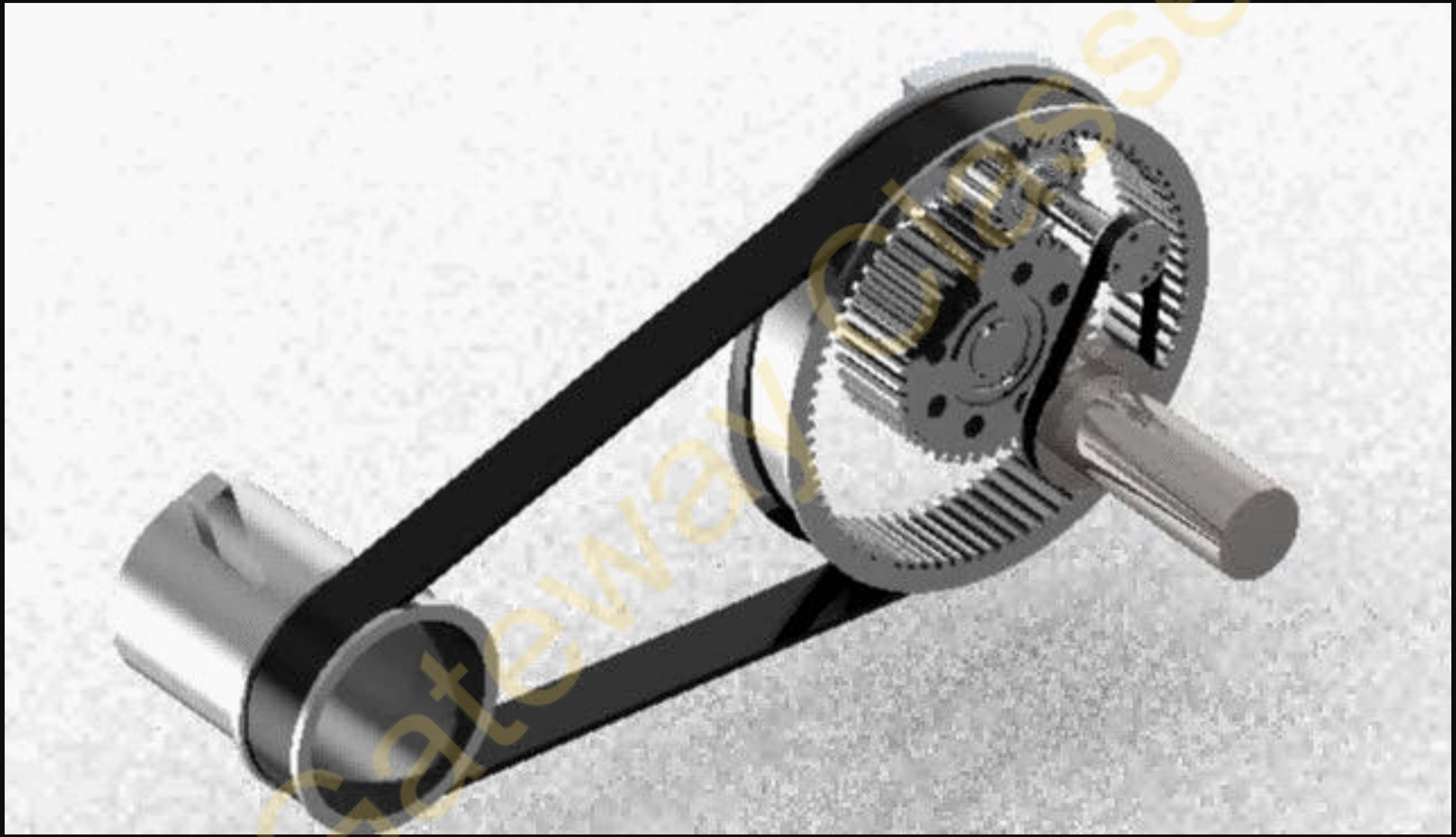
Helical gear : This is a Cylindrical gear. Teeth have helix curve. Helical gear provides more strength, less oscillation and lower noise level compared with Spur gears.

- **Internal gear :** This is a cylindrical gear ring with teeth formed at the inner diameter.

Straight bevel (Miter) gear: Miter gear has shaft angle of 90° and gear ratio of 1:1.

Rack and pinion : „A rack is a gear whose pitch diameter is infinite, resulting in a straight line pitch circle. Used to convert rotary motion to straight line.

(v) Belt Drive



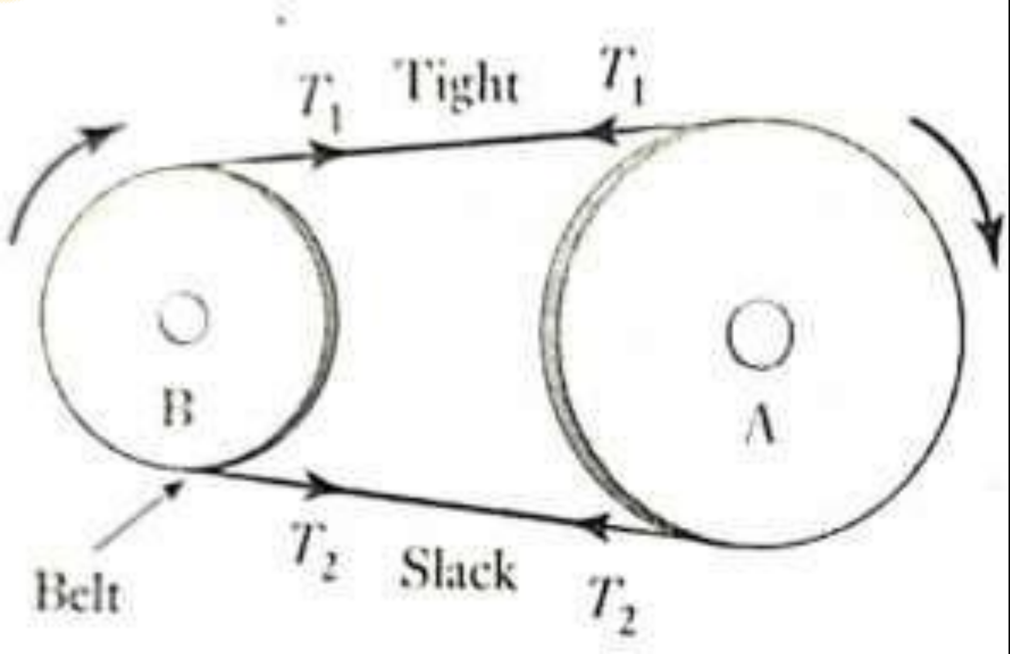
(v) Belt

- Belt drives use the friction that develops between the pulleys attached to the shafts and the belt around the arc of contact in order to transmit a torque.
- The torque is due to the differences in tension that occur in the belt during operation.
- Let T_1 is the tension in the tight side and T_2 is the tension in slack side.

$$\text{torque on A} = (T_1 - T_2)r_A$$

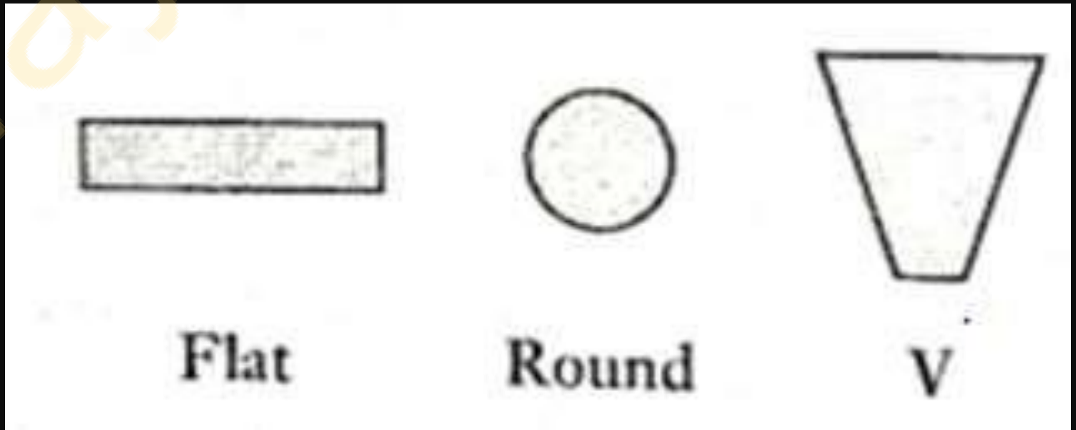
$$\text{torque on B} = (T_1 - T_2)r_B$$

$$\text{power} = (T_1 - T_2)v$$



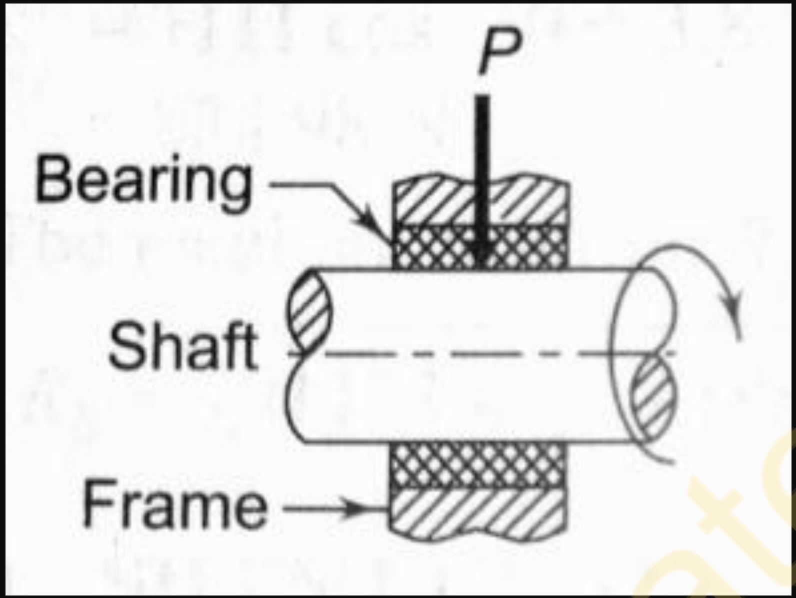
Types of Belt

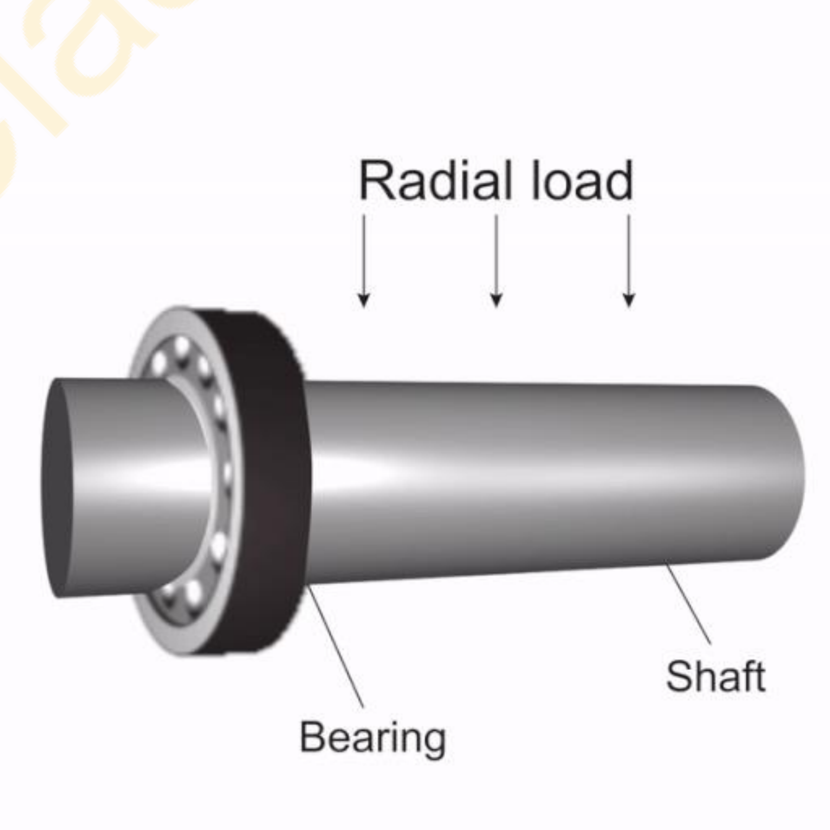
- **Flat:** The belt has a rectangular cross-section and produces less noise. They can transmit power over a long distance between pulley centers
- **Round:** The belt has a circular cross-section and is used with grooved pulleys.
- **V:** V-belts are used with grooved pulleys and are less efficient than flat belts.



(vi) Bearing

A bearing is a mechanical component that supports and reduces friction between moving parts, allowing for smooth and efficient motion within a machine or mechanical system.



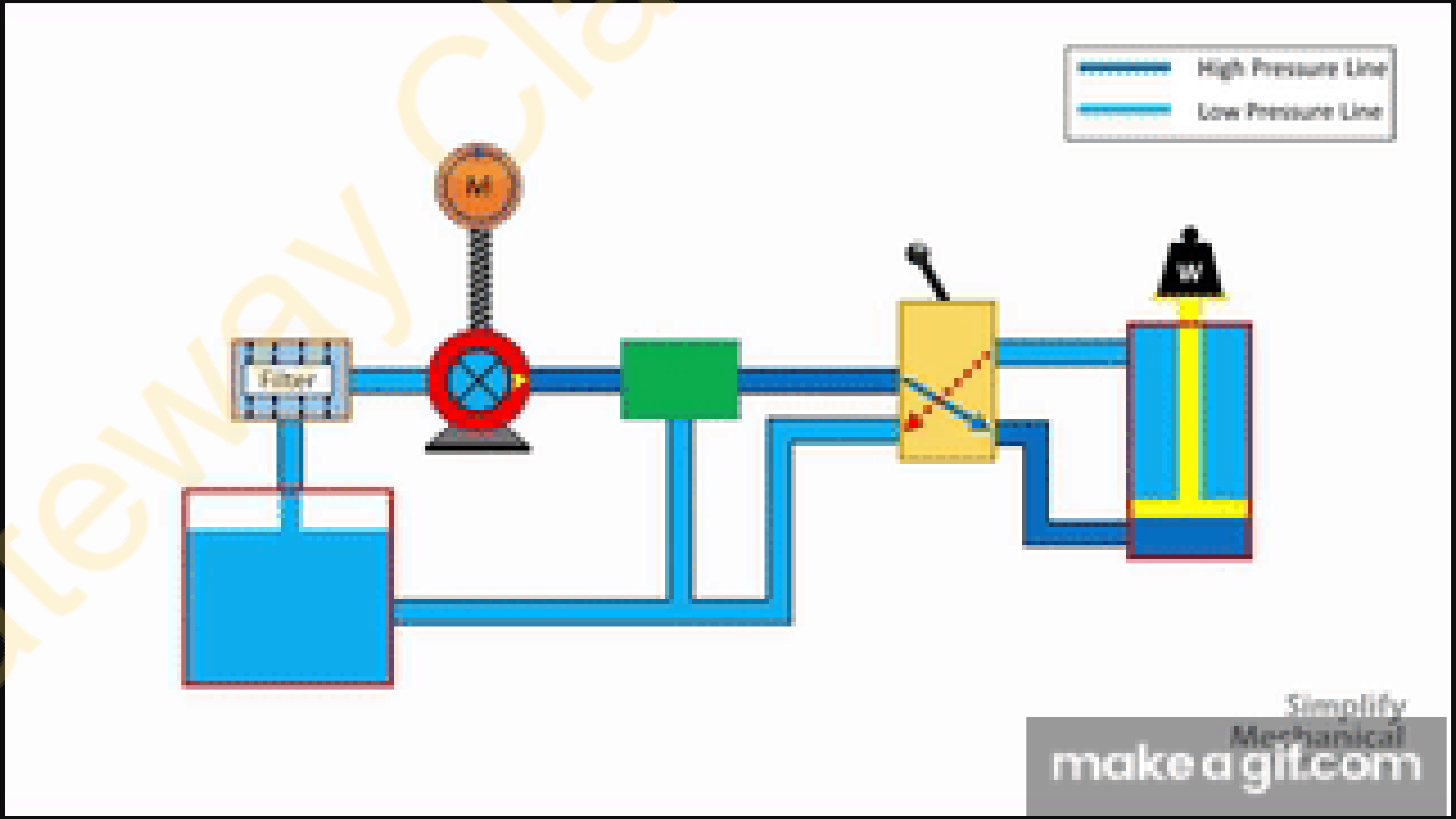


Q.5 Explain Hydraulic and Pneumatic Actuation Systems.

- The controlled movement of parts or a controlled application of force is a common requirement in industries.
- These operations are performed mainly by using electrical machines or diesel, petrol, and steam engines as a prime mover.
- These prime movers can provide various movements to the objects by using mechanical attachments like screw jacks, levers, racks, pinions, etc.
- However, these are **not the only** prime movers. The enclosed fluids (**liquids and gases**) can also be used as prime movers to provide controlled motion and force to the objects.

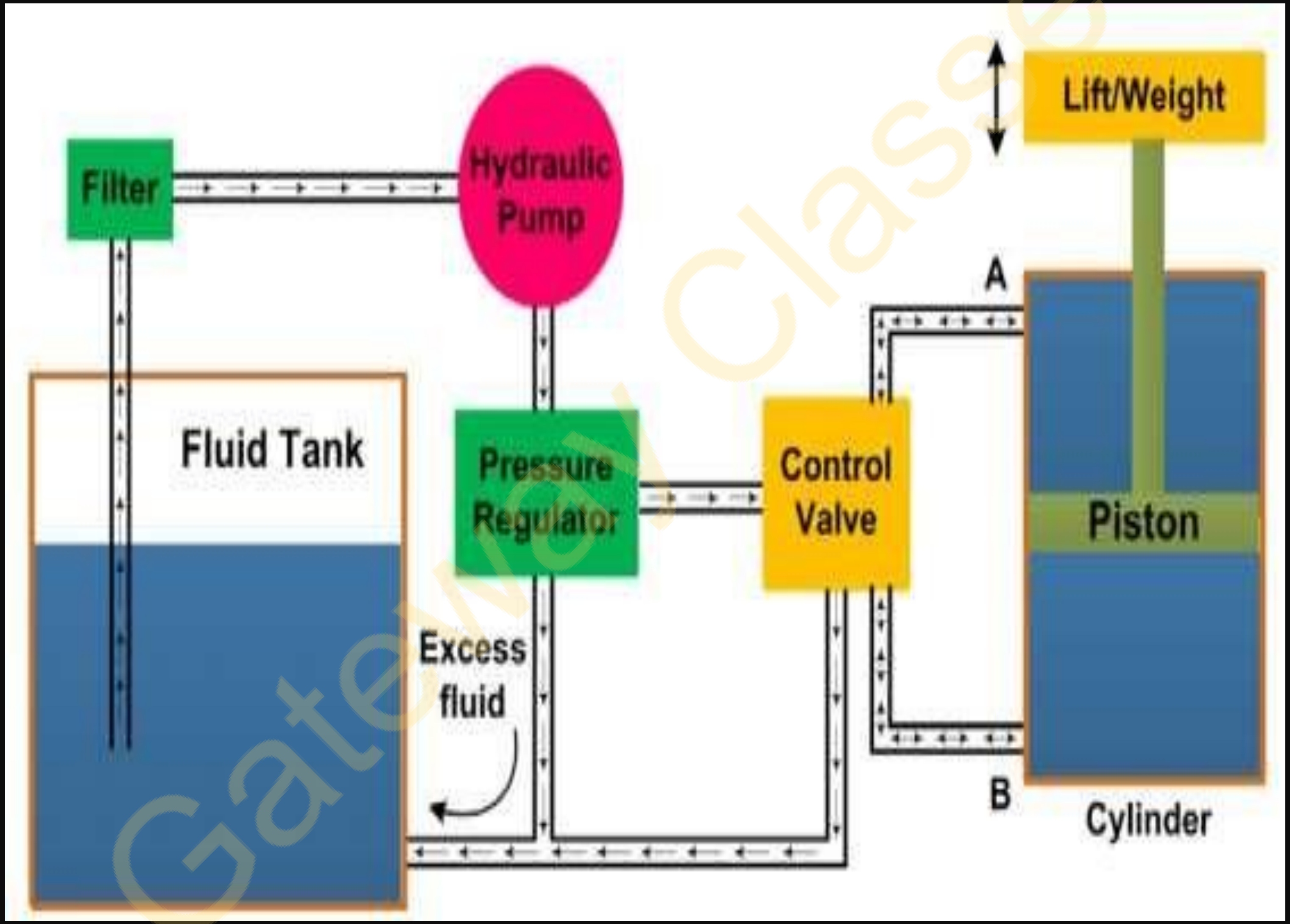
Q.5 Hydraulic and Pneumatic Actuation Systems

- The hydraulic system works on the principle of Pascal’s law which says that the pressure in an enclosed fluid is uniform in all the directions.



6
1

Hydraulic Actuation Systems



[redacted]

The hydraulic system consists a number of parts for its proper functioning. It consists of:

- a movable piston connected to the output shaft in an enclosed cylinder
- storage tank
- filter
- electric pump
- pressure regulator
- control valve
- leakproof closed loop piping.

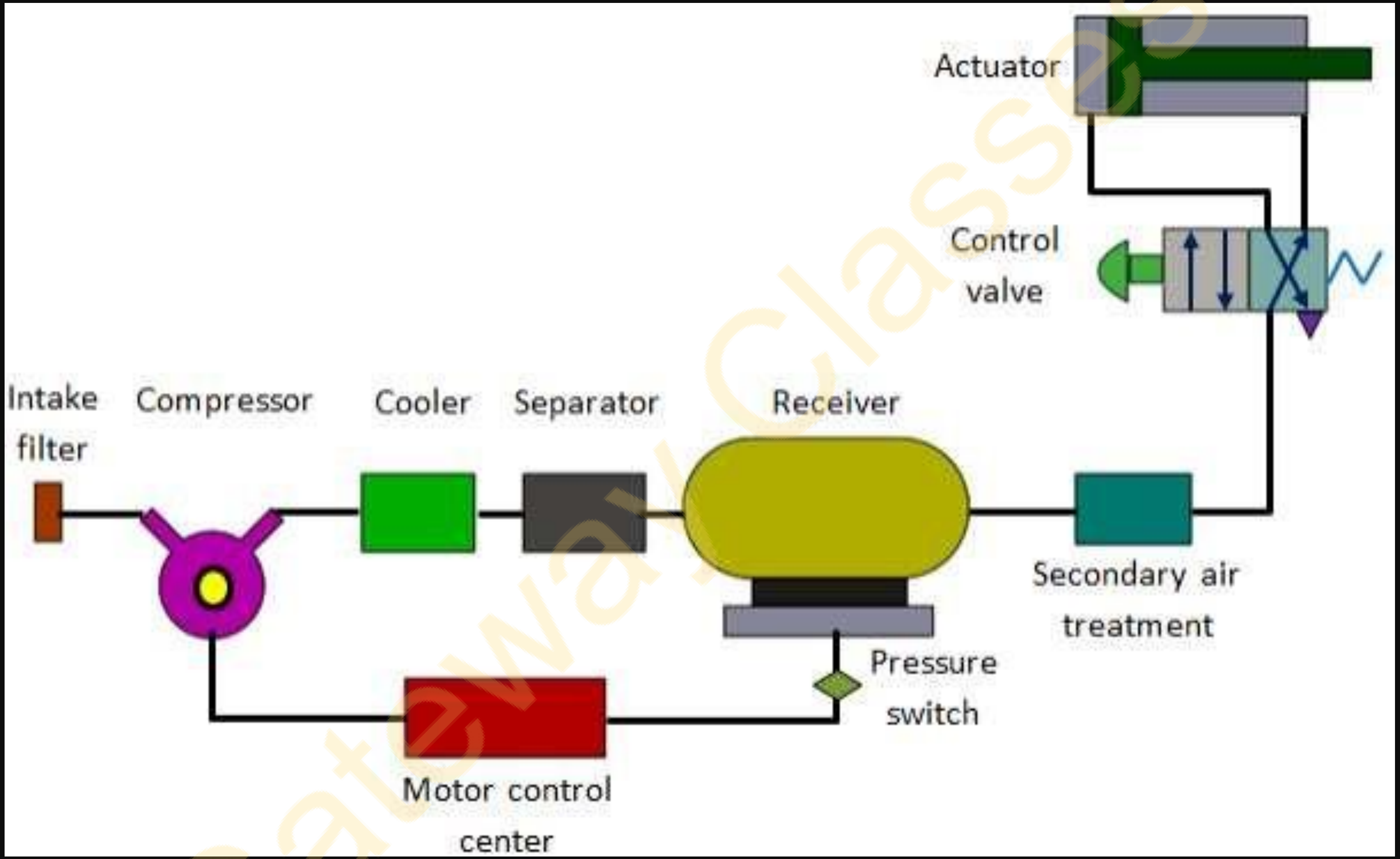
Applications of hydraulic systems

- Construction Equipment
- Manufacturing machinery
- Aerospace systems
- Automotive systems
- Marine applications
- Agricultural machinery
- Material handling equipment
- Power generation
- Civil engineering structures
- Oil and gas industry.

Pneumatic Actuation System.

- Pneumatic technology deals with the study of behavior and applications of **compressed air** in our daily life in general and manufacturing automation in particular.
- Pneumatic systems use **air** as the medium which is abundantly available and can be exhausted into the atmosphere after completion of the assigned task.

Basic Components of Pneumatic System?



Important components of a pneumatic system are shown in fig.

a) **Air filters:** These are used to filter out the contaminants from the air.

b)**Compressor:** Compressed air is generated by using air compressors. Air compressors are either diesel or electrically operated. Based on the requirement of compressed air, suitable capacity compressors may be used.

c)**Air cooler:** During compression operation, air temperature increases. Therefore coolers are used to reduce the temperature of the compressed air.

d) **Dryer:** The water vapor or moisture in the air is separated from the air by using a dryer.

e)Control Valves: Control valves are used to regulate, control, and monitor for control of direction flow, pressure, etc.

f)Air Actuator: Air cylinders and motors are used to obtain the required movements of mechanical elements of a pneumatic system.

g)Electric Motor: Transforms electrical energy into mechanical energy. It is used to drive the compressor.

h)Receiver tank: The compressed air coming from the compressor is stored in the air receiver

Q.6 What do you mean by Valves in hydraulic and pneumatic systems?

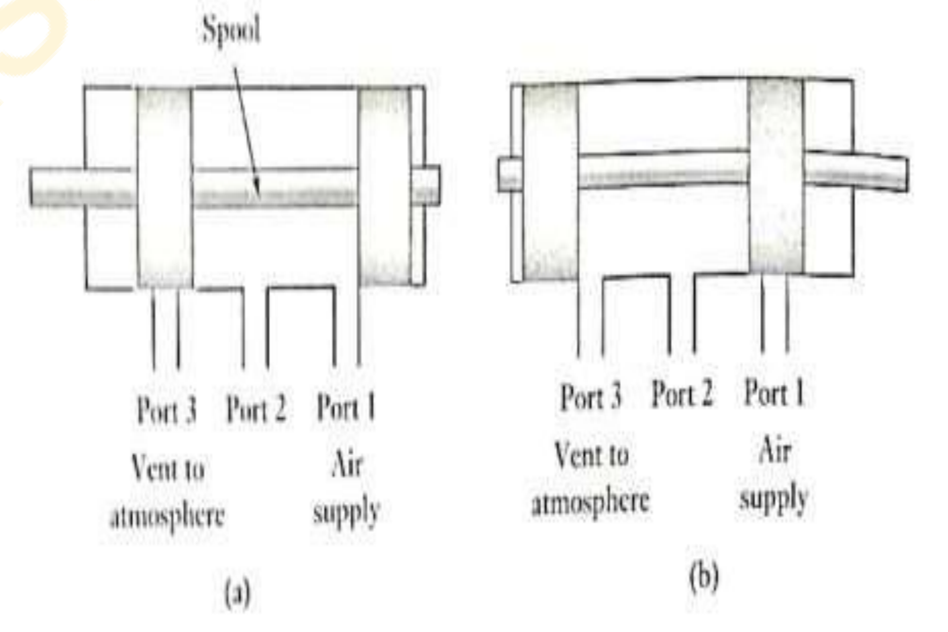
AKTU-2022-23(E) 7 Marks

- In hydraulic and pneumatic systems, valves are mechanical devices that control fluid flow (liquid or gas) within the system.
- Valves are crucial components because they regulate the direction, pressure, and quantity of fluid flow, allowing for precise control and operation of various hydraulic and pneumatic equipment.

Functions and types of valves commonly used in hydraulic and pneumatic systems:

1.Directional Control valves: Valves determine the direction of fluid flow within the system.

- Pneumatic and hydraulic systems use directional control Valves to direct the flow of fluid through a system.
- They do not vary the rate of fluid flow but are either completely open or completely closed i.e. ON/OFF devices.
- They might be activated to switch the fluid flow direction by means of mechanical, electrical, or fluid pressure signals.
- A common type of directional control valve is the spool valve.



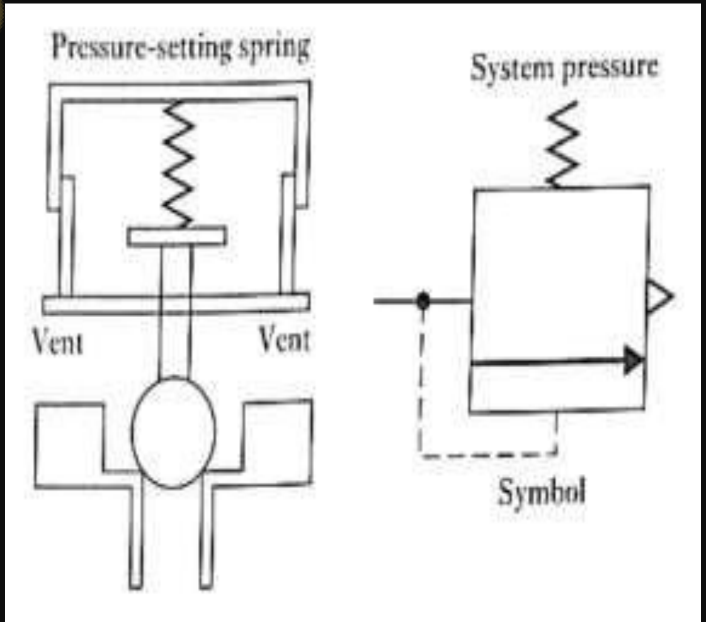
2. Pressure Control Valve: These are used to control the pressure in hydraulic and pneumatic systems.

Pressure – limiting/relief valves

Pressure sequence valves : Sequence valves control the sequence of operations in hydraulic systems, ensuring that certain actions occur in a specific order.

Pressure-Limiting/Relief Valves:

Function: Pressure-limiting valves, also known as relief valves, are designed to limit the maximum pressure in a hydraulic system by diverting excess fluid flow back to the reservoir when the pressure exceeds a predetermined threshold.



Operation: When the pressure in the system exceeds the set limit, the relief valve opens, allowing fluid to bypass the primary circuit and return to the reservoir, thus preventing overpressure and potential damage to the system components.

Applications: They are commonly used in hydraulic systems to protect pumps, actuators, and other components from damage due to overpressure conditions.

3. Flow Control Valve: Valves regulate the flow rate of fluid passing through the system, controlling the speed of actuators or cylinders.

4. **Check Valves:** These valves allow fluid to flow in one direction only, preventing backflow and maintaining system integrity.

5. Sequence Valves: Sequence valves control the sequence of operations in hydraulic systems, ensuring that certain actions occur in a specific order.

Q.7 What do you mean by Actuators? Explain Linear and Rotary actuators.

AKTU-2022-23(E) 2Marks

- Actuators are output devices which convert energy from pressurized **hydraulic oil** or **compressed air** into the required type of action or motion.
- In general, hydraulic or pneumatic systems are used for gripping and/or moving operations in industry. These operations are carried out by using actuators.
- In general actuators can be classified into two types.

1. **Linear actuators:** These devices convert hydraulic/pneumatic energy into linear motion.**(Ex-cylinder)**
2. **Rotary actuators:** These devices convert hydraulic/pneumatic energy into rotary motion. **(Ex-Gear motor)**

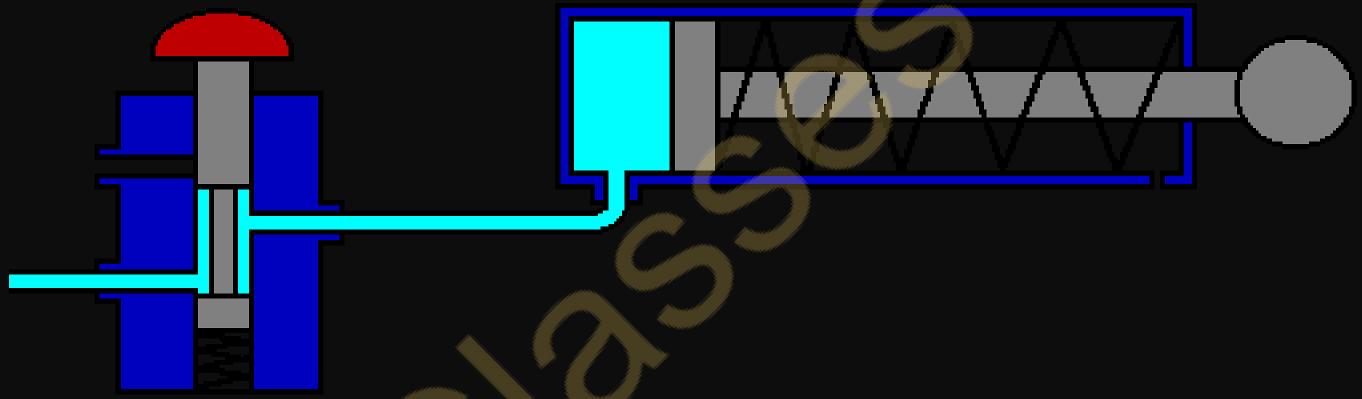
The construction of hydraulic and pneumatic linear actuators is similar.

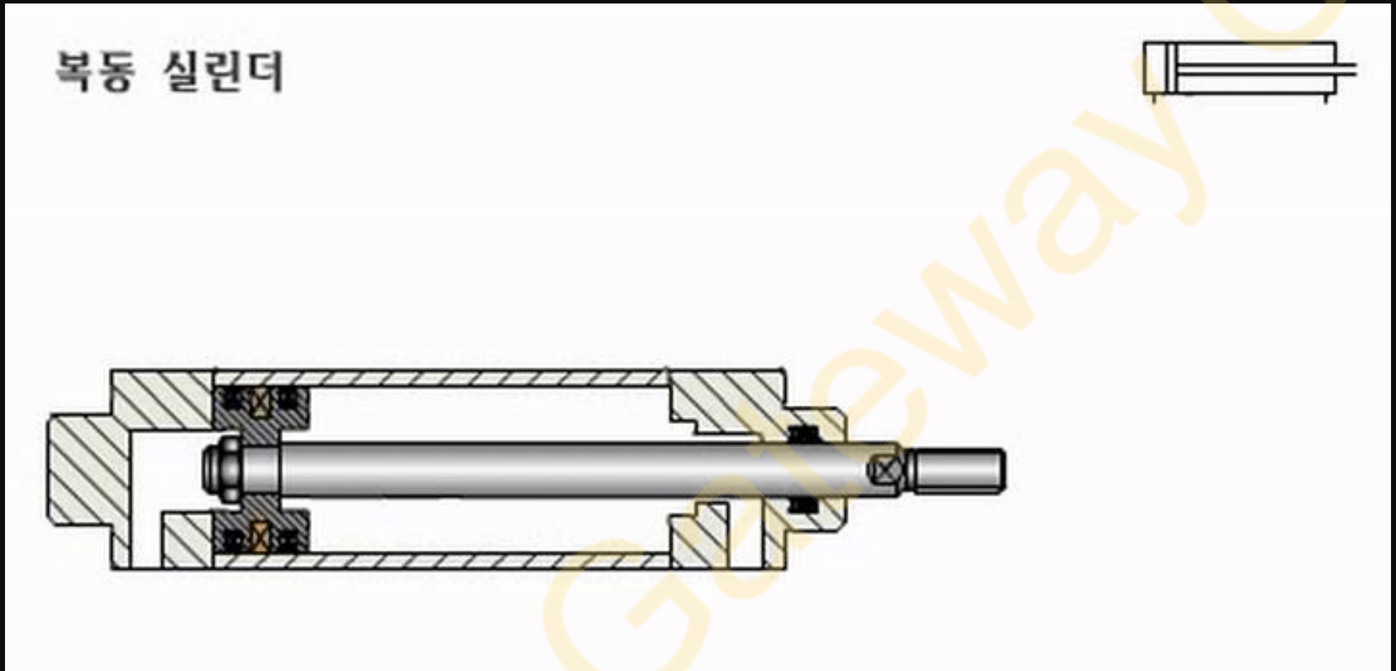
However, they differ in their operating pressure ranges.

The typical pressure of hydraulic cylinders is about 100 bar and of pneumatic system is around 10 bar.

Linear Actuator :

Single-acting cylinder.



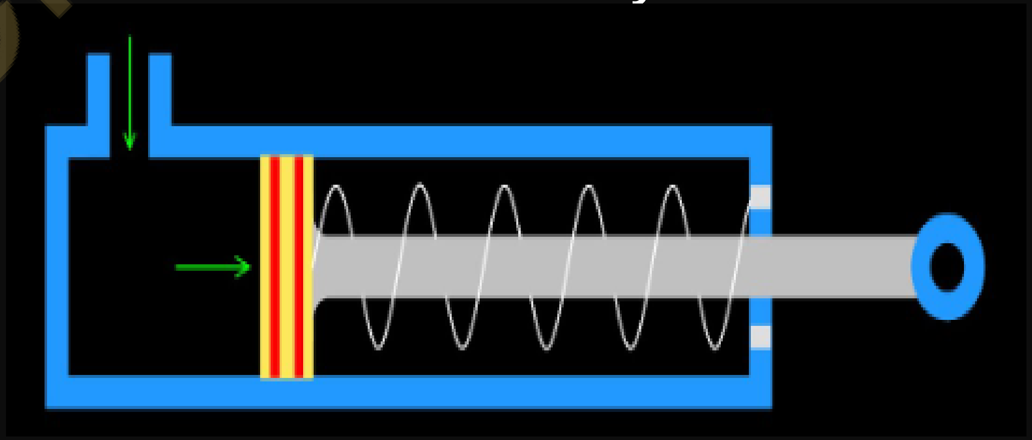


Linear Actuator :

Double-acting cylinder.

Linear Actuator :

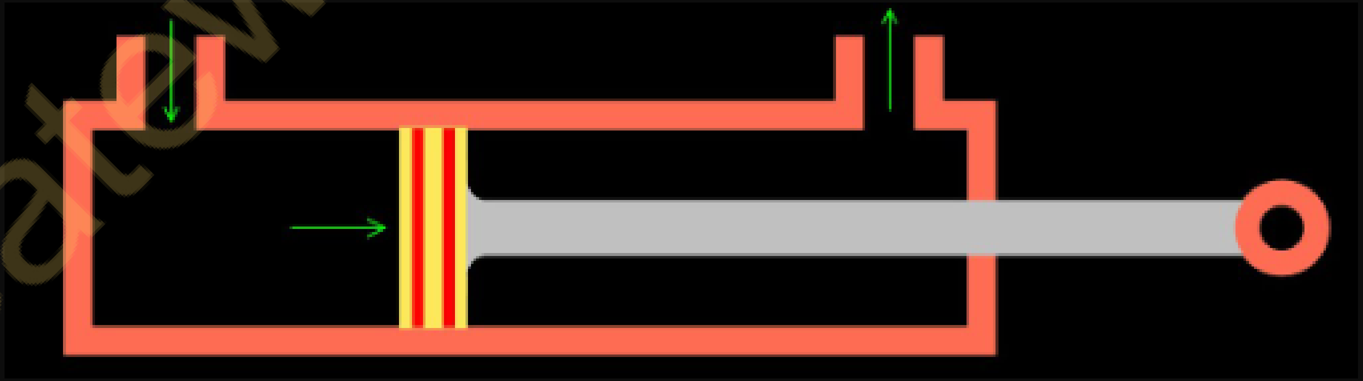
- These cylinders produce work in one direction of motion hence they are named as single-acting cylinders.
- The figure shows the construction of a Single-acting cylinder.
- The compressed air pushes the piston located in the cylindrical barrel causing the desired motion.



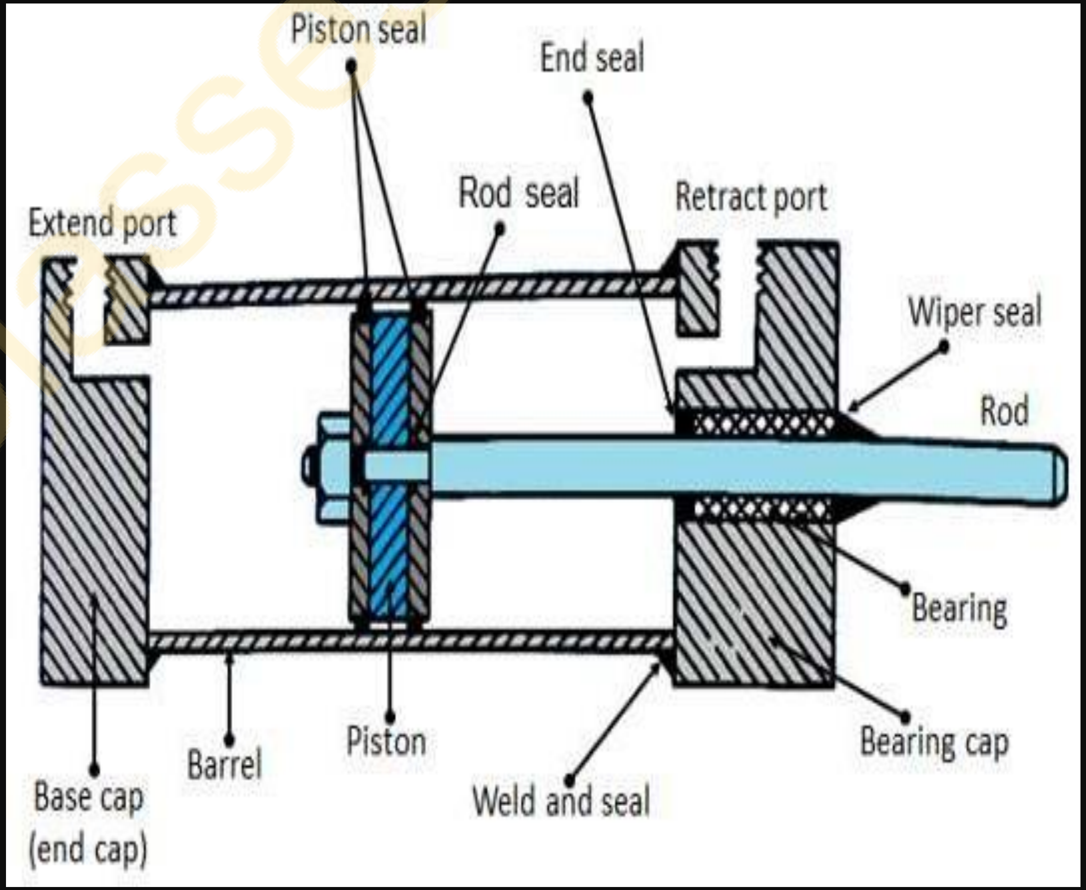
- The return stroke takes place by the action of a spring.
- Generally, the spring is provided on the rod side of the cylinder.

Linear Actuator: Double-acting cylinder.

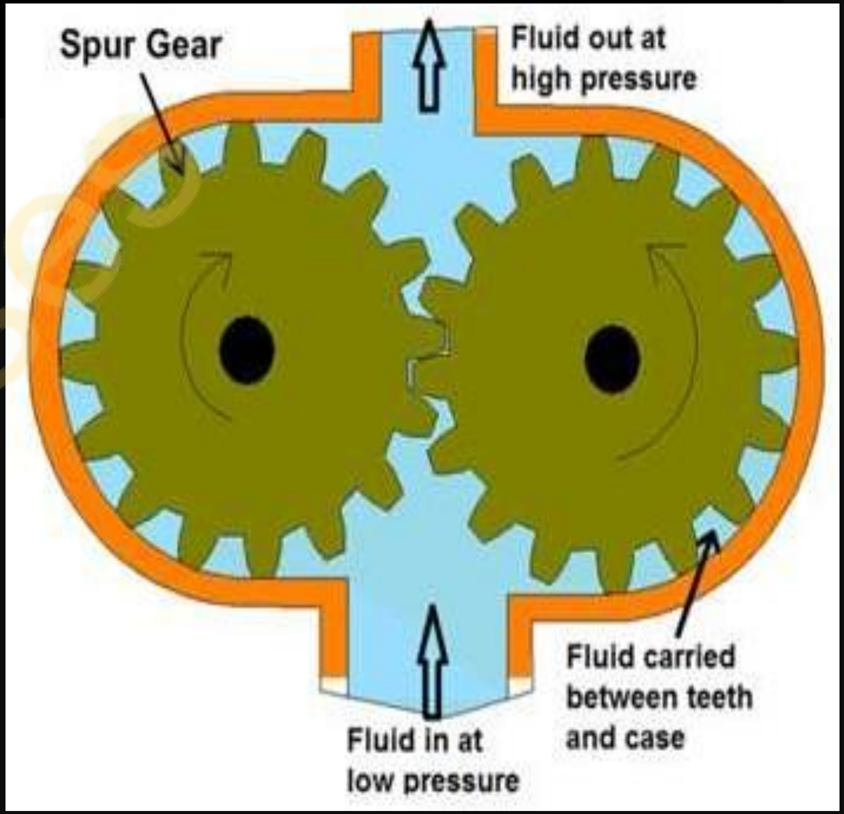
- The main parts of a hydraulic double-acting cylinder are the piston, piston rod, cylinder tube, and end caps.
- As shown in the fig. the piston rod is connected to the piston head and the other end extends out of the cylinder.
- The piston divides the cylinder into two chambers.
- The seals prevent the leakage of oil between these two chambers.
- The cylindrical tube is fitted with end caps.



- The pressurized oil, air enters the cylinder chamber through the ports provided.
- In the rod end cover plate, a wiper seal is provided to prevent the leakage of oil and entry of the contaminants into the cylinder.
- The piston seal prevents metal to metal contact and wear of piston head and the tube. These seals are replaceable.
- End cushioning is also provided to prevent the impact with end caps.



Rotary Rotary Actuators



- Rotary actuators convert the energy of pressurized fluid into rotary motion. Rotary actuators are similar to electric motors but are run on hydraulic or pneumatic power.
- It consists of two intermeshing gears inside a housing with one gear attached to the drive shaft.
- Figure shows a schematic diagram of Gear
  - motor.
- The air enters from the inlet, causes the rotation of the meshing gear due to the difference in the pressure, and produces the torque.
- The air exits from the exhaust port.
- Gear motors tend to leak at low speeds and, hence are generally used for medium speed applications.

AKTU : B.Tech (I,II,III-Year)

S.N.	Full Courses available in application		
1	CS, IT & CS Allied	COMBO (Odd + Even) SEM	COMBO Odd/Even SEM
2	EC & EC Allied		
3	ME & ME Allied		
4	EE & EE Allied		

Thank You

Thank you